Arba Minch University
Institute of Technology
Faculty of Electrical and Computer Engineering
Electronic Communication Engineering Stream

"COMPUTATIONAL METHODS"

(ECEg – 3042)

Compiled by:

Mr. Firew Ayele

COMPUTATIONAL METHODS

(ECEg - 3042)

CHAPTER TWO

SOLUTION OF NON – LINEAR EQUATIONS

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- Years ago, you learned to use the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad eq\ (a)$$

to solve:

$$f(x) = ax^2 + bx + c \qquad eq\ (b)$$

- The values calculated with Eq. (a) are called the "roots" of Eq. (b).
- They represent the values of x that make Eq. (b) equal to zero.
- Thus, we can define the root of an equation as the value of x that makes $f(x) = 0$.
- For this reason, roots are sometimes called the zeros of the equation.

Cont …

- Although the quadratic formula is handy for solving Eq. (b), there are many other functions for which the root cannot be determined so easily.
- For these cases, the numerical methods described in this chapter provide efficient means to obtain the answer.
- For example, even an apparently simple function such as:

$$f(x) = e^{-x} - x$$

- Cannot be solved analytically.
- In such instances, the only alternative is an approximate solution technique.

Cont …

- One method to obtain an approximate solution is to plot the function and determine where it crosses the x axis.
- Which is called **Graphical technique**.
- This point, which represents the x value for which $f(x) = 0$, is the root.
- Although graphical methods are useful for obtaining rough estimates of roots, they are limited because of their lack of precision.
- An alternative approach is to use **trial and error**.
- This "technique" consists of guessing a value of x and evaluating whether $f(x)$ is zero.

Cont …

- If not (as is almost always the case), another guess is made, and $f(x)$ is again evaluated to determine whether the new value provides a better estimate of the root.
- The process is repeated until a guess is obtained that results in an $f(x)$ that is close to zero.

Mathematical Background

- By definition, a function given by $y = f(x)$ is algebraic if it can be expressed in the form:

$$f_n y^n + f_{n-1} y^{n-1} + \cdots + f_1 y + f_o = 0$$

Where $f_i =$ an i^{th}- order polynomial in x.

- Polynomials are a simple class of algebraic functions that are represented generally by:

$$f_i(x) = a_o + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

Where n = the order of the polynomial and $a's$ = constants.

Cont …

- Some specific examples are:

$$f_2(x) = 1 - 2.37x + 7.5x^2 \quad and \quad f_6(x) = 5x^2 - x^3 + 7x^6$$

- A transcendental function is one that is non – algebraic.
- These include trigonometric, exponential, logarithmic, and other, less familiar, functions.
- Examples are:

$$f(x) = \ln x^2 - 1 \quad and \quad f(x) = e^{-0.2x}\sin(3x - 0.5)$$

- Roots of equations may be either real or complex.

Bracketing Methods/ Close Methods

- This chapter on roots of equations deals with methods that exploit the fact that a function typically changes sign in the vicinity of a root.
- These techniques are called bracketing methods because two initial guesses for the root are required.
- As the name implies, these guesses must **"bracket"**, or be on either side of, the root.
- The particular methods described herein employ different strategies to systematically reduce the width of the bracket and, hence, home in on the correct answer.

Cont …

- As introduction to these techniques, we will briefly discuss graphical methods for depicting functions and their roots.
- Beyond their utility for providing rough guesses, graphical techniques are also useful for visualizing the properties of the functions and the behavior of the various numerical methods.

Graphical Method

- A simple method for obtaining an estimate of the root of the equation $f(x) = 0$ is to make a plot of the function and observe where it crosses the x axis.
- This point, which represents the x value for which $f(x) = 0$, provides a rough approximation of the root.

Example 1: The graphical approach

- **Problem Statement**: Use the graphical approach to determine the drag coefficient 'c' needed for a parachutist of mass $m = 68.1\ kg$ to have a velocity of $40\ m/s$ after free-falling for time $t = 10\ s$.

Cont …

- Note: The acceleration due to gravity is $9.8\,m/s^2$ and for the parachutist's velocity:

$$v = \frac{gm}{c}\left(1 - e^{-\left(\frac{c}{m}\right)t}\right)$$

$$f(c) = \frac{gm}{c}\left(1 - e^{-\left(\frac{c}{m}\right)t}\right) - v$$

- The value of c that makes $f(c) = 0$ is, therefore, the root of the equation.
- This value also represents the drag coefficient that solves the design problem.

Cont …

Solution: Substituting these values yield

$$f(c) = \frac{gm}{c}\left(1 - e^{-\left(\frac{c}{m}\right)t}\right) - v = \frac{9.8 * 68.1}{c}\left(1 - e^{-\left(\frac{c}{68.1}\right)*10}\right) - 40$$

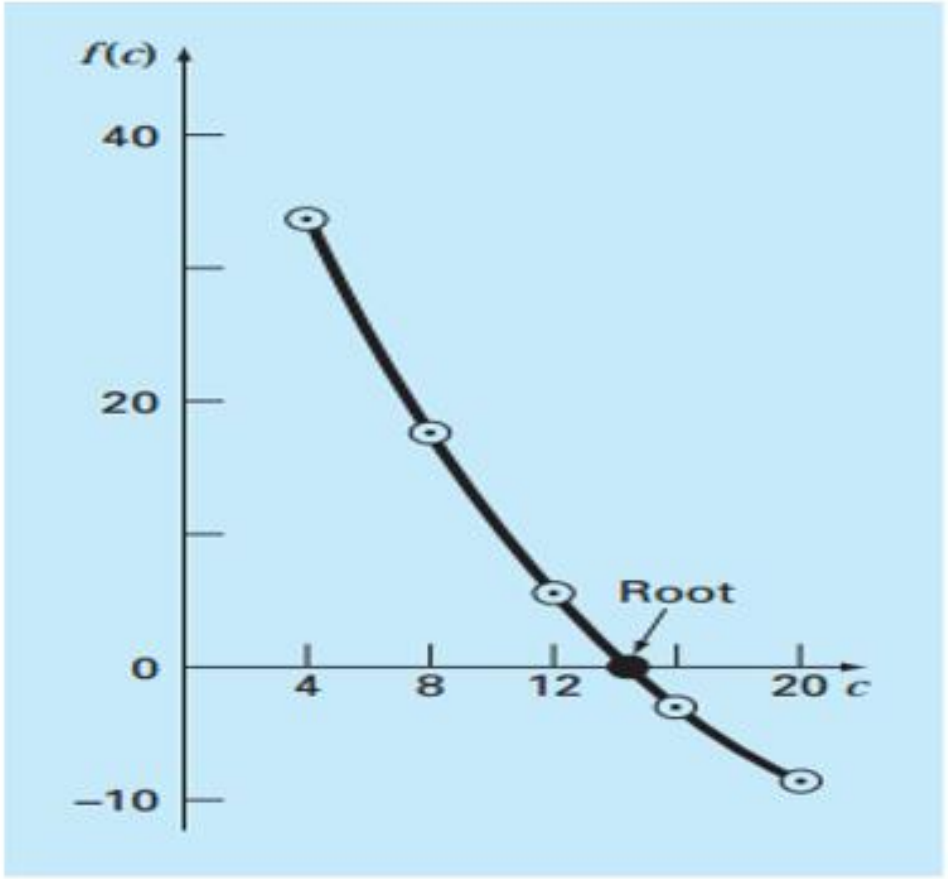
Or

$$f(c) = \frac{667.38}{c}(1 - e^{-0.146843c}) - 40$$

- Various values of 'c' can be substituted into the right – hand side of this equation to compute:

c	f(c)
4	34.115
8	17.653
12	6.067
16	−2.269
20	−8.401

The graphical approach for determining the roots of an equation.



Cont …

- These points are plotted in Figure above.
- The resulting curve crosses the c axis between 12 and 16.
- Visual inspection of the plot provides a rough estimate of the root of 14.75.
- The validity of the graphical estimate can be checked by substituting it into equation for $f(c)$ to yield:

$$f(14.75) = \frac{667.38}{14.75}(1 - e^{-0.146843*14.75}) - 40 = 0.059$$

Cont …

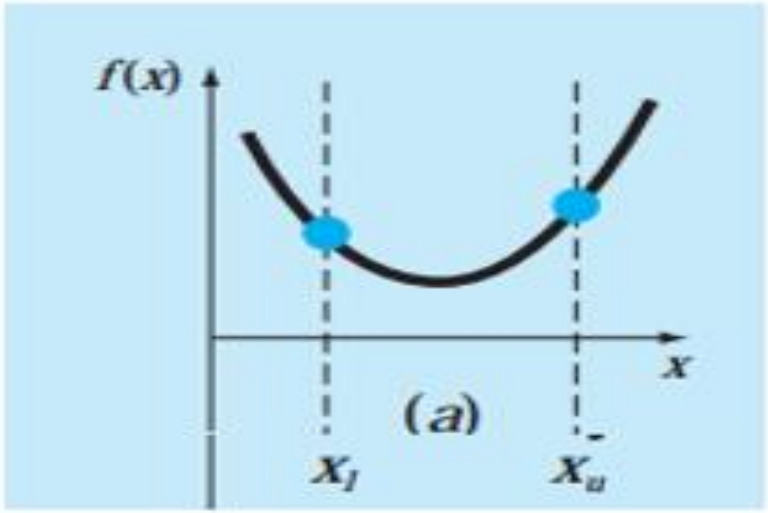
- Which is close to zero.
- It can also be checked by substituting it into equation for 'v' along with the parameter values from this example to give:

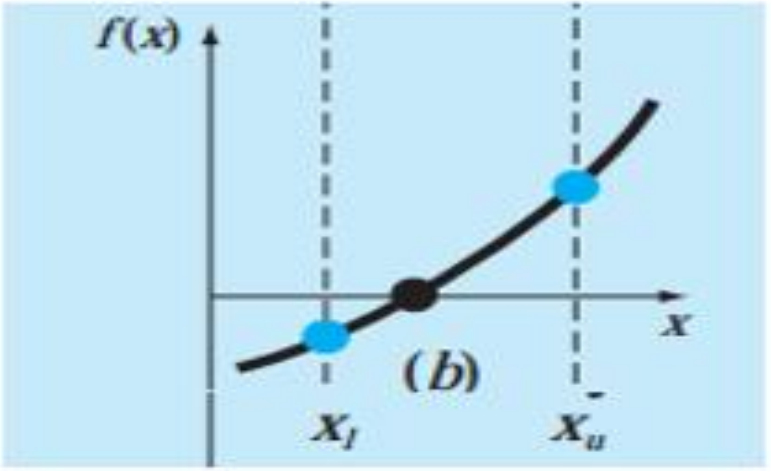
$$v = \frac{gm}{c}\left(1 - e^{-\left(\frac{c}{m}\right)t}\right) = \frac{9.8 * 68.1}{14.75}\left(1 - e^{-\left(\frac{14.75}{68.1}\right)*10}\right) = 40.059$$

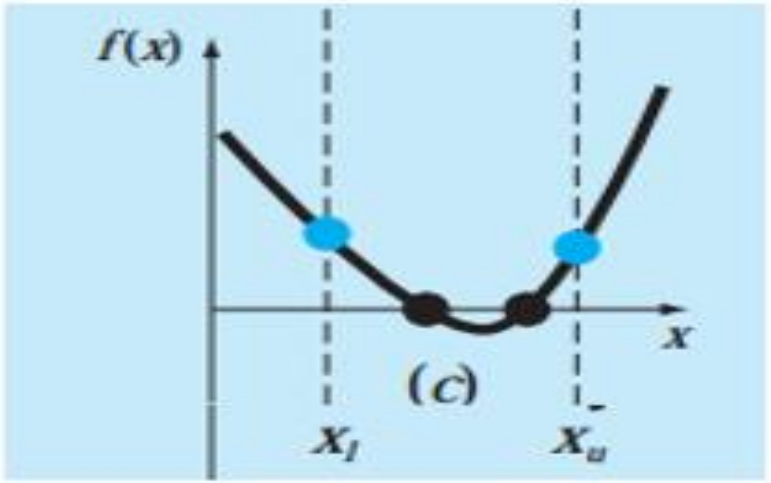
- Which is very close to the desired fall velocity of 40 m/s.
- Graphical techniques are of limited practical value because they are not precise.
- However, graphical methods can be utilized to obtain rough estimates of roots.

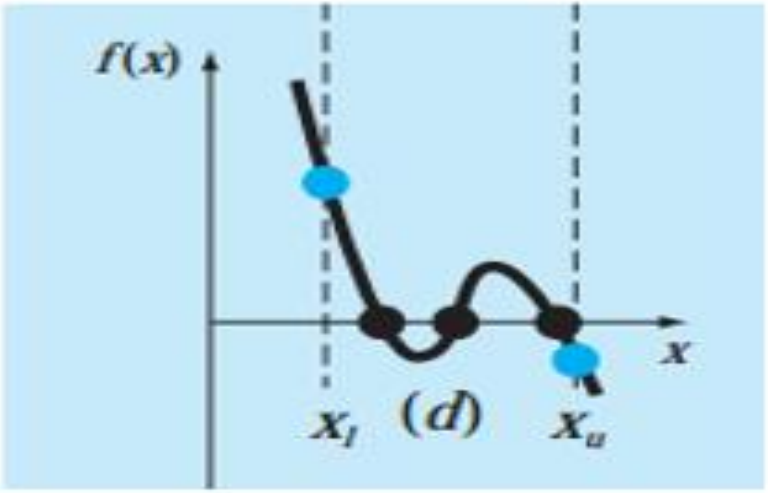
Cont …

- These estimates can be employed as starting guesses for numerical methods.
- Aside from providing rough estimates of the root, graphical interpretations are important tools for understanding the properties of the functions and anticipating the pitfalls/drawbacks of the numerical methods.
- For example, Figure below shows a number of ways in which roots can occur (or be absent) in an interval prescribed by a lower bound x_l and an upper bound x_u.


f(x)
x
(a)
x_l
x_u


f(x)
x
(b)
x_l
x_u


f(x)
x
(c)
x_l
x_u

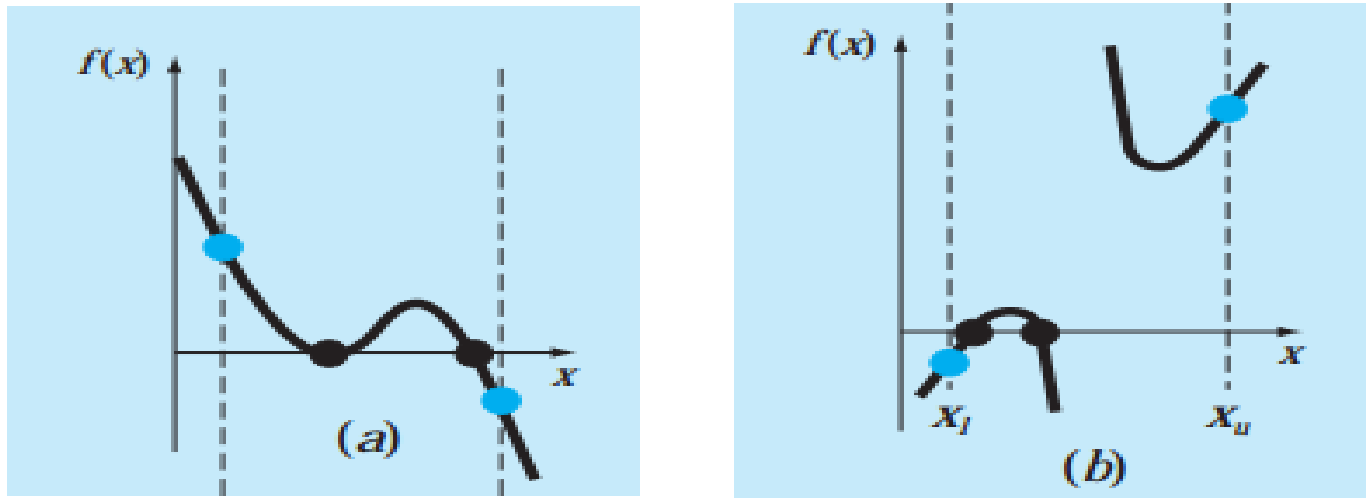

f(x)
x
x_l
(d)
x_u

Cont …

- Figure (b) depicts the case where a single root is bracketed by negative and positive values of $f(x)$.
- However, Fig. (d), where $f(x_l)$ and $f(x_u)$ are also on opposite sides of the x axis, shows three roots occurring within the interval.
- In general, if $f(xl)$ and $f(xu)$ have opposite signs, there are an odd number of roots in the interval.
- As indicated by Fig. (a) and (c), if $f(x_l)$ and $f(x_u)$ have the same sign, there are either no roots or an even number of roots between the values.

Cont …

- Although these generalizations are usually true, there are cases where they do not hold.
- For example, functions that are tangential to the x axis (Fig. '*a*' below) and discontinuous functions (Fig. '*b*' below) can violate these principles.



- An example of a function that is tangential to the axis is the cubic equation:

$$f(x) = (x - 2)(x - 2)(x - 4)$$

- Notice that $x = 2$ makes two terms in this polynomial equal to zero.
- Mathematically, $x = 2$ is called a multiple root.
- The existence of cases of the type depicted in Figure above makes it difficult to develop general computer algorithms guaranteed to locate all the roots in an interval.
- However, when used in conjunction with graphical approaches, the methods described in the following sections are extremely useful for solving many roots of equations problems.

The Bisection Method

- When applying the graphical technique above, we have observed that $f(x)$ changed sign on opposite sides of the root.
- In general, if $f(x)$ is real and continuous in the interval from bound x_l to x_u and $f(x_l)$ and $f(x_u)$ have opposite signs, that is,

$$f(x_l)f(x_u) < 0$$

- Then there is at least one real root between x_l and x_u.
- Incremental search methods capitalize on this observation by locating an interval where the function changes sign.

Cont …

- Then the location of the sign change (and consequently, the root) is identified more precisely by dividing the interval into a number of subintervals.
- Each of these subintervals is searched to locate the sign change.
- The process is repeated and the root estimate refined by dividing the subintervals into finer increments.
- The **bisection method**, which is alternatively called binary chopping, interval halving, or Bolzano's method, is one type of incremental search method in which the interval is always divided in half.

Cont …

- If a function changes sign over an interval, the function value at the midpoint is evaluated.
- The location of the root is then determined as lying at the midpoint of the subinterval within which the sign change occurs.
- The process is repeated to obtain refined estimates.
- A simple algorithm for the bisection calculation is listed in steps below and a graphical depiction of the method is provided in Figure below.

Cont …

Step 1: Choose lower x_l and upper x_u guesses for the root such that the function changes sign over the interval.

- This can be checked by ensuring that

$$f(x_l)f(x_u) < 0$$

Step 2: An estimate of the root x_r is determined by:

$$x_r = \frac{x_l + x_u}{2}$$

Cont …

Step 3: Make the following evaluations to determine in which subinterval the root lies:

a) If $f(x_l)f(x_r) < 0$, the root lies in the lower subinterval. Therefore, set $x_u = x_r$ and return to step 2.

b) If $f(x_l)f(x_r) > 0$, the root lies in the upper subinterval. Therefore, set $x_l = x_r$ and return to step 2.

c) If $f(x_l)f(x_r) = 0$, the root equals x_r. Terminate the computation.

- The following example goes through the actual computations involved in the method.

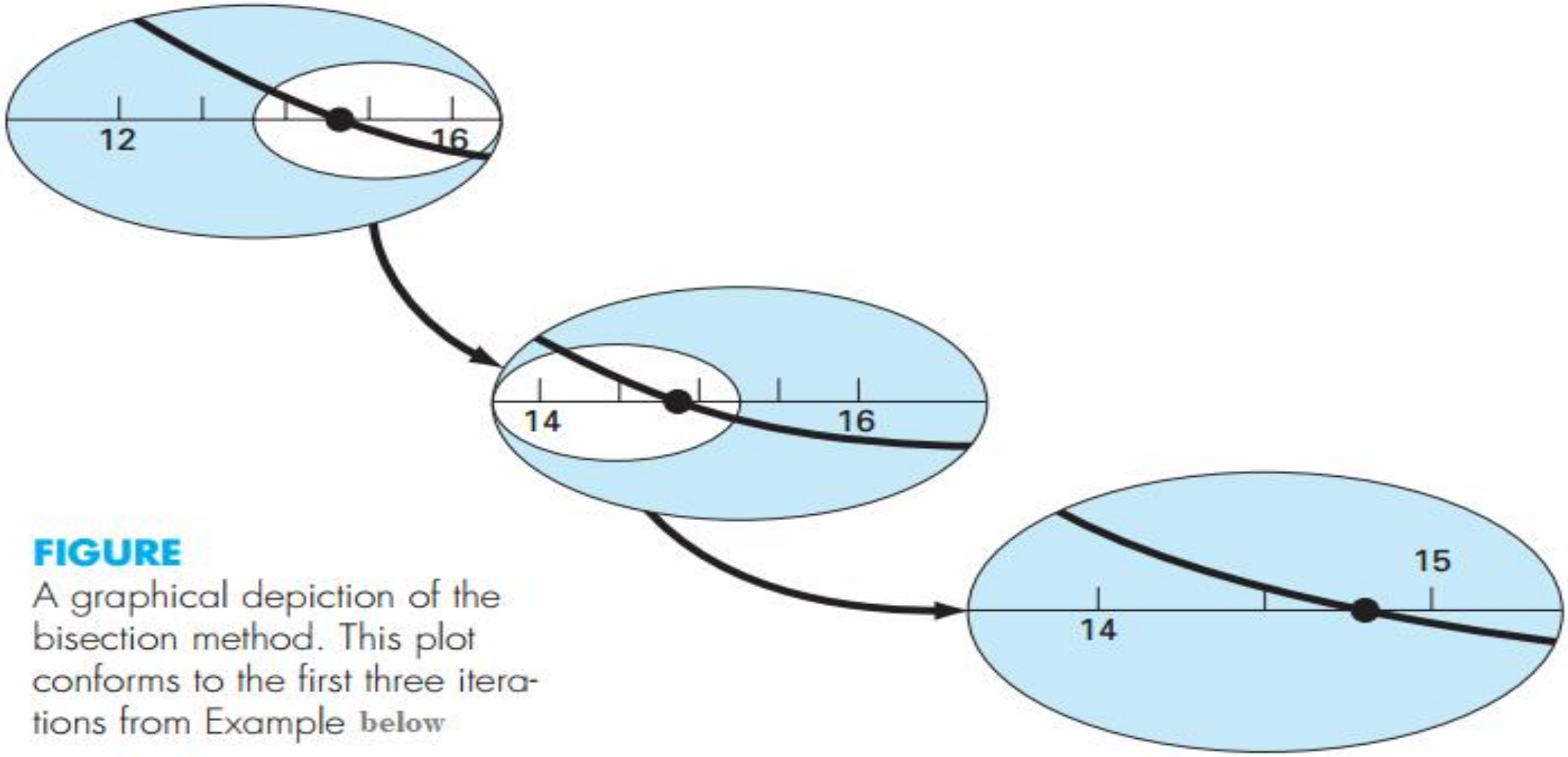


FIGURE
A graphical depiction of the bisection method. This plot conforms to the first three iterations from Example below

Example: Bisection

Problem Statement:

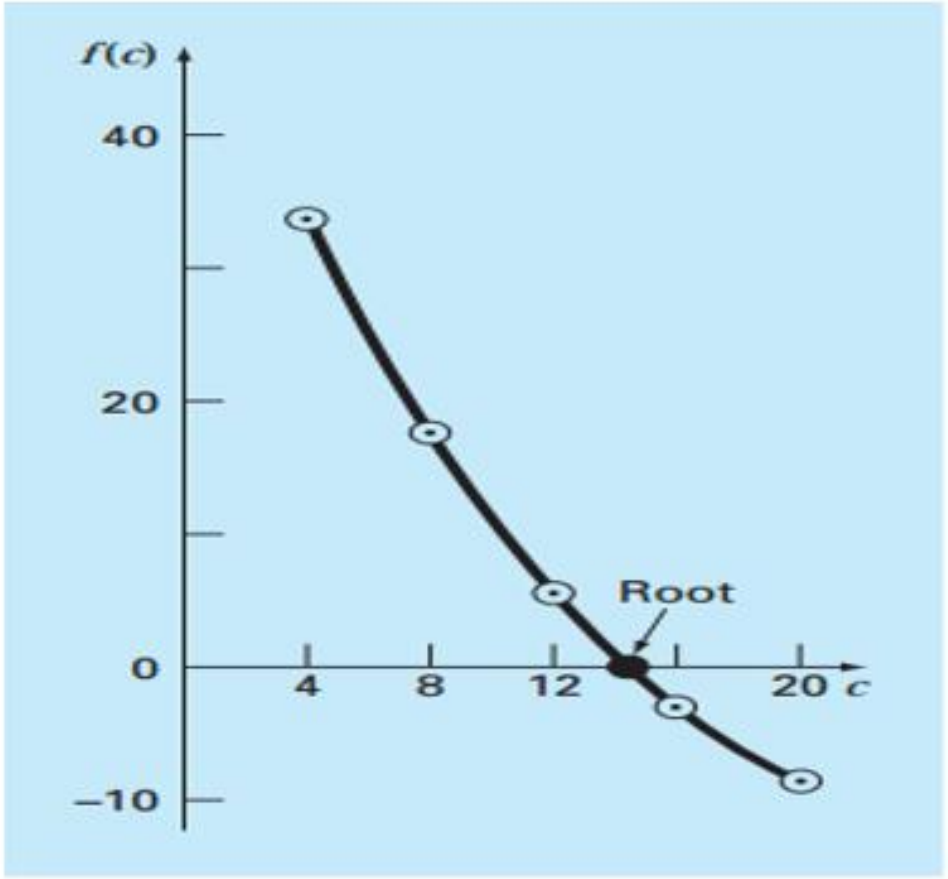
- Use bisection to solve the same problem approached graphically above (Parachutist).

Solution:

- The first step in bisection is to guess two values of the unknown (in the present problem, c) that give values for $f(c)$ with different signs from Example 1, we can see that the function changes sign between values of 12 and 16.

c	f(c)
4	34.115
8	17.653
12	6.067
16	−2.269
20	−8.401

The graphical approach for determining the roots of an equation.



Cont …

- Therefore, the initial estimate of the root x_r lies at the midpoint of the interval

$$x_r = \frac{12 + 16}{2} = 14$$

- This estimate represents a true percent relative error of $\varepsilon_t = 5.3\%$ (note that the true value of the root is 14.7802).
- Next we compute the product of the function value at the lower bound and at the midpoint:

$$f(12) * f(14) = 6.067 * 1.569 = 9.517$$

Cont …

- Which is greater than zero, and hence no sign change occurs between the lower bound and the midpoint.
- Consequently, the root must be located between 14 and 16.
- Therefore, we create a new interval by redefining the lower bound as 14 and determining a revised root estimate as:

$$x_r = \frac{14 + 16}{2} = 15$$

- Which represents a true percent error of $\varepsilon_t = 1.3\%$.

Cont …

- The process can be repeated to obtain refined estimates.
- For example,

$$f(14) * f(15) = 1.569 * (-0.425) = -0.666$$

- Therefore, the root is between 14 and 15.
- The upper bound is redefined as 15, and the root estimate for the third iteration is calculated as:

$$x_r = \frac{14 + 15}{2} = 14.5$$

- Which represents a percent relative error of $\varepsilon_t = 1.9\ \%$.

Cont …

- The method can be repeated until the result is accurate enough to satisfy your needs.
- In the previous example, you may have noticed that the true error does not decrease with each iteration.
- However, the interval within which the root is located is halved with each step in the process.

Termination Criteria and Error Estimates

- We ended the above example with the statement that the method could be continued to obtain a refined estimate of the root.

- We must now develop an objective criterion for deciding when to terminate the method.
- An initial suggestion might be to end the calculation when the true error falls below some pre – specified level.
- For instance, in Example above, the relative error dropped from 5.3 to 1.9 percent during the course of the computation.
- We might decide that we should terminate when the error drops below, say, 0.1 percent.

Cont …

- This strategy is flawed because the error estimates in the example were based on knowledge of the true root of the function.
- This would not be the case in an actual situation because there would be no point in using the method if we already knew the root.
- Therefore, we require an error estimate that is not contingent on foreknowledge of the root.
- As developed previously, an approximate percent relative error ε_a can be calculated, as:

$$\varepsilon_a = \left| \frac{x_r^{new} - x_r^{old}}{x_r^{new}} \right| * 100\%$$

Cont …

- Where x_r^{new}: is the root for the present iteration and x_r^{old} is the root from the previous iteration.
- The absolute value is used because we are usually concerned with the magnitude of ε_a rather than with its sign.
- When ε_a becomes less than a pre – specified stopping criterion ε_s, the computation is terminated. i.e.;

$$|\varepsilon_a| < \varepsilon_s$$

Example: Error Estimates for Bisection

- **Problem Statement :** Continue the above Example until the approximate error falls below a stopping criterion of $\varepsilon_s = 0.5\%$.
- Use the equation below to compute the errors.

$$\varepsilon_a = \left| \frac{x_r^{new} - x_r^{old}}{x_r^{new}} \right| * 100\%$$

Solution: The results of the first two iterations for the above Example were 14 and 15. Substituting these values into the equation yields:

$$\varepsilon_a = \left| \frac{15 - 14}{15} \right| * 100\% = 6.667\%$$

Cont …

- Recall that the true percent relative error for the root estimate of 15 was 1.5%. Therefore, ε_a is greater than ε_t.
- This behavior is manifested for the other iterations:

Iteration	x_l	x_u	x_r	ε_a (%)	ε_t (%)
1	12	16	14		5.279
2	14	16	15	6.667	1.487
3	14	15	14.5	3.448	1.896
4	14.5	15	14.75	1.695	0.204
5	14.75	15	14.875	0.840	0.641
6	14.75	14.875	14.8125	0.422	0.219

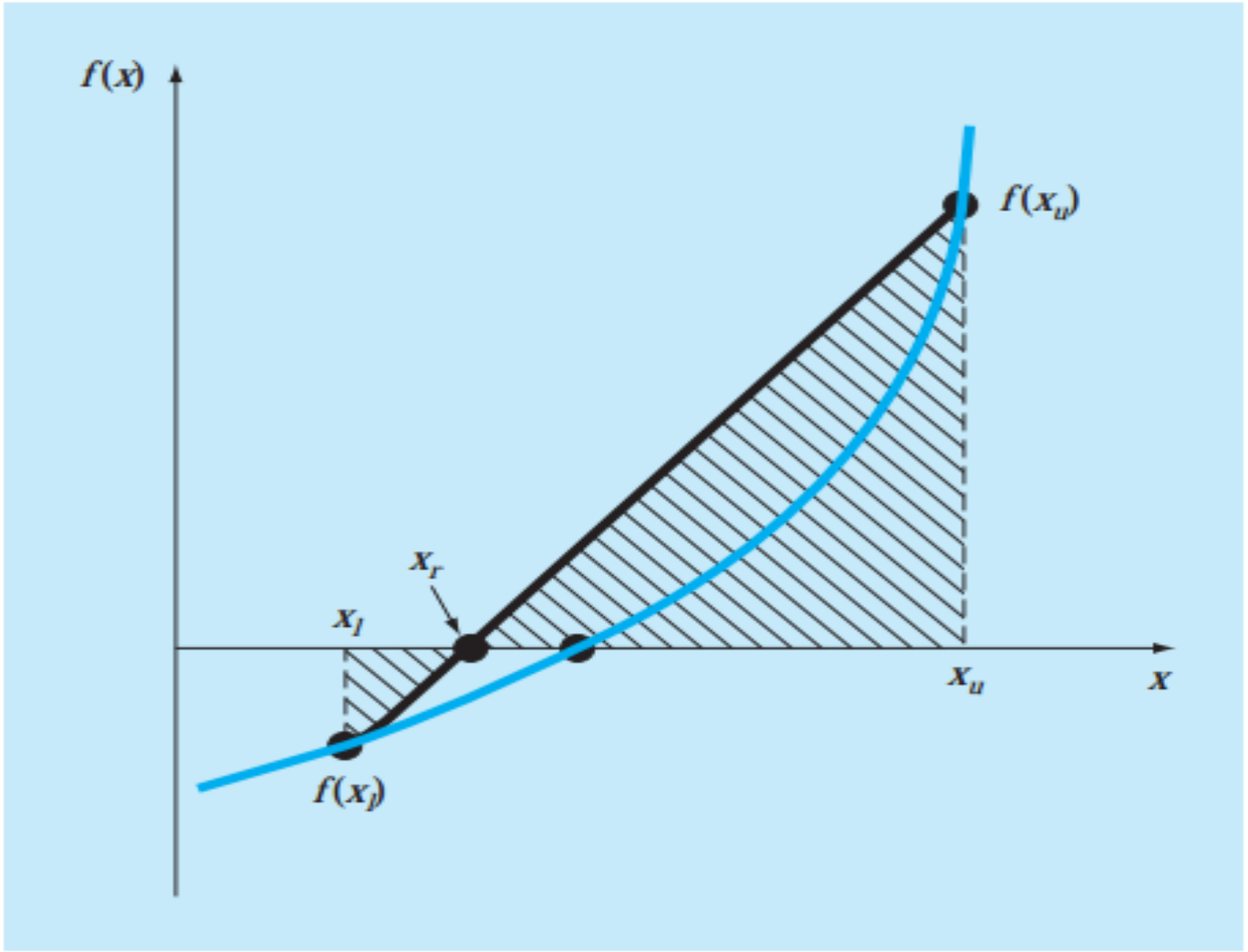
Cont …

- Thus, after six iterations ε_a finally falls below $\varepsilon_s = 0.5\%$, and the computation can be terminated.
- These results are summarized above.
- The "ragged" nature of the true error is due to the fact that, for bisection, the true root can lie anywhere within the bracketing interval.
- The true and approximate errors are far apart when the interval happens to be centered on the true root.
- They are close when the true root falls at either end of the interval.

The False – Position Method

- Although bisection is a perfectly valid technique for determining roots, its "brute force" approach is relatively inefficient.
- False position is an alternative based on a graphical insight.
- A shortcoming of the bisection method is that, in dividing the interval from $x_l \; to \; x_u$ into equal halves, no account is taken of the magnitudes of $f(x_l) \; and \; f(x_u)$.
- For example; If $f(x_l)$ is much closer to zero than $f(x_u)$, it is likely that the root is closer to x_l than to x_u (Fig. below).

FIGURE
A graphical depiction of the method of false position. Similar triangles used to derive the formula for the method are shaded.



Cont …

- An alternative method that exploits this graphical insight is to join $f(x_l)\ and\ f(x_u)$ by a straight line.
- The intersection of this line with the x axis represents an improved estimate of the root.
- The fact that the replacement of the curve by a straight line gives a "false position" of the root is the origin of the name, method of false position, or in Latin, **regula falsi**.
- It is also called the linear interpolation method.

Cont …

- Using similar triangles (Fig. above), the intersection of the straight line with the x axis can be estimated as:

$$\frac{f(x_l)}{x_r - x_l} = \frac{f(x_u)}{x_r - x_u}$$

- Re – arranging for x_r:

$$x_r = \frac{x_u f(x_l) - x_l f(x_u)}{f(x_l) - f(x_u)}$$

- Expanding the equation:

$$x_r = \frac{x_u f(x_l)}{f(x_l) - f(x_u)} - \frac{x_l f(x_u)}{f(x_l) - f(x_u)}$$

Cont …

- Then adding and subtracting x_u on the right – hand side:

$$x_r = x_u + \frac{x_u f(x_l)}{f(x_l) - f(x_u)} - x_u - \frac{x_l f(x_u)}{f(x_l) - f(x_u)}$$

- Collecting the two middle terms yields:

$$x_r = x_u + \frac{x_u f(x_u)}{f(x_l) - f(x_u)} - \frac{x_l f(x_u)}{f(x_l) - f(x_u)}$$

- Finally;

$$x_r = x_u - \frac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$$

- This is the **false-position formula.**

Cont …

- The value of x_r, computed with false position formula, then replaces whichever of the two initial guesses, x_l or x_u, yields a function value with the same sign as $f(x_r)$.
- In this way, the values of x_l and x_u always bracket the true root.
- The process is repeated until the root is estimated adequately.
- The algorithm is identical to the one for bisection with the exception that false position formula is used for step 2.
- In addition, the same stopping criterion is used to terminate the computation.

Example: False Position

- **Problem Statement:** Use the false – position method to determine the root of the same equation investigated in Example above (Parachutist).
- **Solution:** As in Example above, initiate the computation with guesses of $x_l = 12 \; and \; x_u = 16$.
- First iteration:

$$x_l = 12 \qquad f(x_l) = 6.0699$$

$$x_u = 16 \qquad f(x_u) = -2.2688$$

$$x_r = 16 - \frac{-2.2688(12 - 16)}{6.0699 - (-2.2688)} = 14.9113$$

Cont …

- Which has a true relative error of 0.89%.
- Second iteration:

$$f(x_l)f(x_r) = -1.5426$$

Therefore, the root lies in the first subinterval, and x_r becomes the upper limit for the next iteration, $x_u = 14.9113$.

$$x_l = 12 \qquad f(x_l) = 6.0699$$

$$x_u = 14.9113 \qquad f(x_u) = -2.2543$$

$$x_r = 14.9113 - \frac{-2.2543(12 - 14.9113)}{6.0699 - (-2.2543)} = 14.7942$$

Cont …

- Which has true and approximate relative errors of 0.09% and 0.79%.
- Additional iterations can be performed to refine the estimate of the roots.
- Note how the error for false position decreases much faster than for bisection because of the more efficient scheme for root location in the false – position method.

Pitfalls of the False Position Method

- Although the false – position method would seem to always be the bracketing method of preference, there are cases where it performs poorly.
- In fact, as in the following example, there are certain cases where bisection yields superior results.

Example: A Case Where Bisection is Preferable to False Position

- **Problem Statement:** Use bisection and false position to locate the root of:

$$f(x) = x^{10} - 1 \qquad \text{between } x = 0 \text{ and } 1.3$$

Solution: Using bisection, the results can be summarized as: Cont …

Iteration	x_l	x_u	x_r	ε_a (%)	ε_t (%)
1	0	1.3	0.65	100.0	35
2	0.65	1.3	0.975	33.3	2.5
3	0.975	1.3	1.1375	14.3	13.8
4	0.975	1.1375	1.05625	7.7	5.6
5	0.975	1.05625	1.015625	4.0	1.6

- Thus, after five iterations, the true error is reduced to less than 2 percent.
- For false position, a very different outcome is obtained:

Iteration	x_l	x_u	x_r	ε_a (%)	ε_t (%)
1	0	1.3	0.09430		90.6
2	0.09430	1.3	0.18176	48.1	81.8
3	0.18176	1.3	0.26287	30.9	73.7
4	0.26287	1.3	0.33811	22.3	66.2
5	0.33811	1.3	0.40788	17.1	59.2

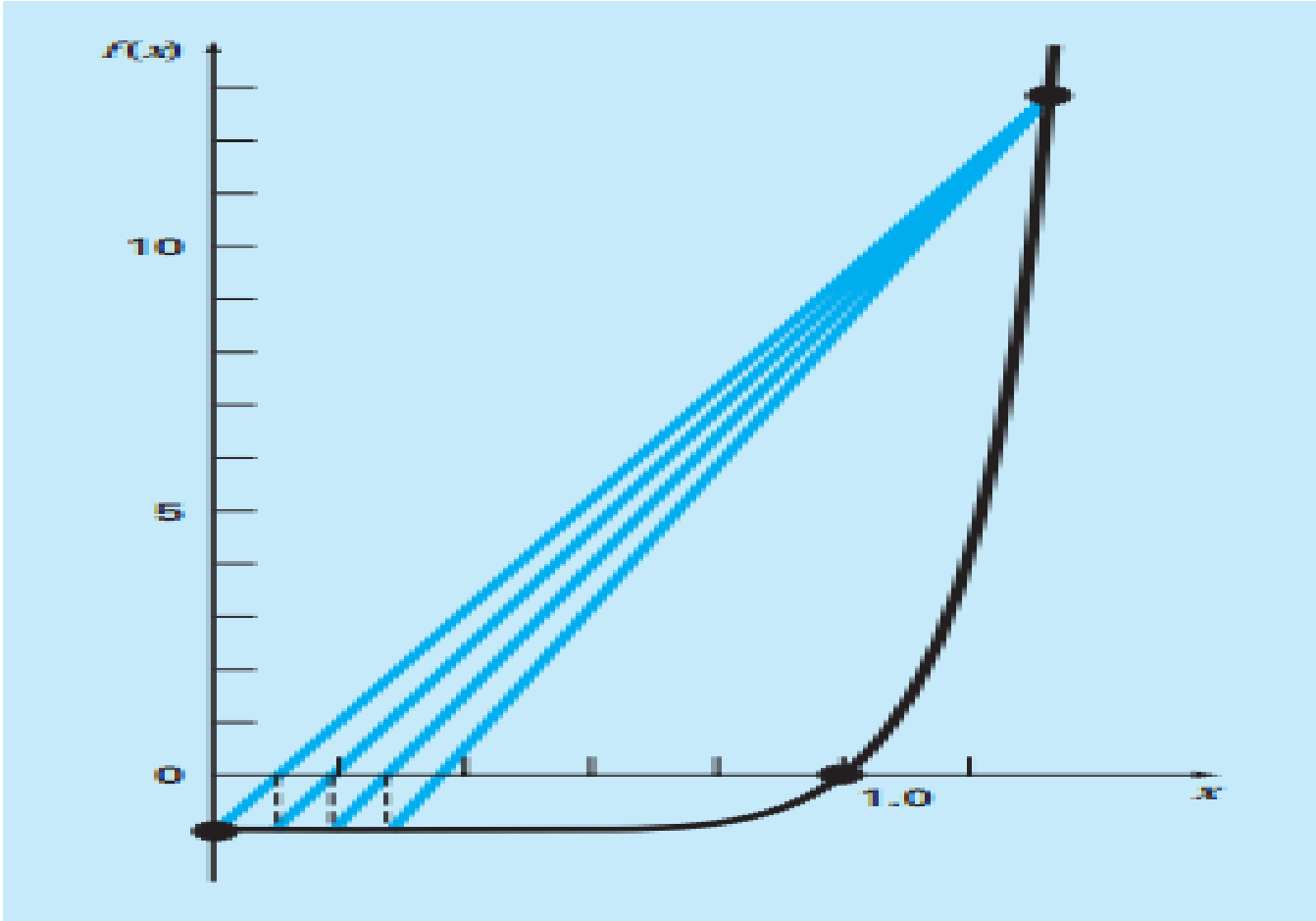


FIGURE
Plot of $f(x) = x^{10} - 1$, illustrating slow convergence of the false-position method.

Cont …

- After five iterations, the true error has only been reduced to about 59 percent.

- In addition, note that $\varepsilon_a < \varepsilon_t$. Thus, the approximate error is misleading.
- Insight into these results can be gained by examining a plot of the function.
- As in figure for the plot of $f(x)$, the curve violates the premise upon which false position was based – that is, if $f(x_l)$ is much closer to zero than $f(x_u)$, then the root is closer to x_l than to x_u.
- Because of the shape of the present function, the opposite is true.

Cont …

- The forgoing/previous example illustrates that blanket generalizations regarding root – location methods are usually not possible.
- Although a method such as false position is often superior to bisection, there are invariably cases that violate this general conclusion.
- Therefore, in addition to using equation for ε_a, the results should always be checked by substituting the root estimate into the original equation and determining whether the result is close to zero.
- Such a check should be incorporated into all computer programs for root location.

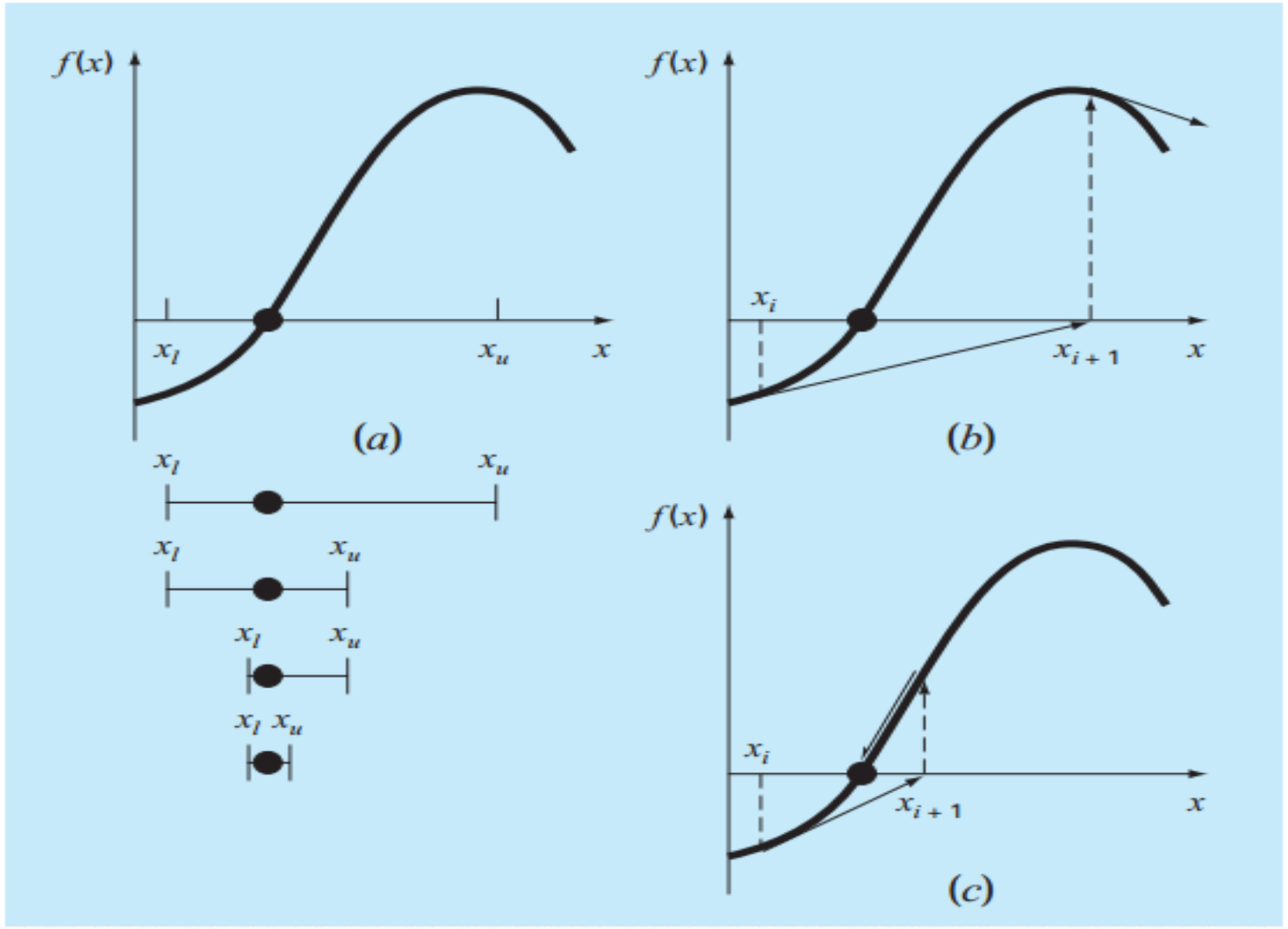
Cont …

- The example also illustrates a major weakness of the false – position method: its one sidedness.
- That is, as iterations are proceeding, one of the bracketing points will tend to stay fixed.
- This can lead to poor convergence, particularly for functions with significant curvature.

Iterative Methods/ Open Methods

- For the bracketing methods discussed previously, the root is located within an interval prescribed by a lower and an upper bound.
- Repeated application of these methods always results in closer estimates of the true value of the root.
- Such methods are said to be convergent because they move closer to the truth as the computation progresses (Fig. a below).
- In contrast, the open methods described in this section are based on formulas that require only a single starting value of x or two starting values that do not necessarily bracket the root.

FIGURE
Graphical depiction of the fundamental difference between the (*a*) bracketing and (*b*) and (*c*) open methods for root location. In (*a*), which is the bisection method, the root is constrained within the interval prescribed by x_l and x_u. In contrast, for the open method depicted in (*b*) and (*c*), a formula is used to project from x_i to x_{i+1} in an iterative fashion. Thus, the method can either (*b*) diverge or (*c*) converge rapidly, depending on the value of the initial guess.



Cont …

- As such, they sometimes diverge or move away from the true root as the computation progresses (Fig. b).
- However, when the open methods converge (Fig. c), they usually do so much more quickly than the bracketing methods.
- We will begin our discussion of open techniques with a simple version that is useful for illustrating their general form and also for demonstrating the concept of convergence.

Simple Fixed - Point Iteration

- As mentioned above, open methods employ a formula to predict the root.

- Such a formula can be developed for simple fixed – point iteration (or, as it is also called, one – point iteration or successive substitution) by re – arranging the function $f(x) = 0$ so that x is on the left – hand side of the equation:

$$x = g(x)$$

- This transformation can be accomplished either by algebraic manipulation or by simply adding x to both sides of the original equation.

Cont …

- For example,

$$x^2 - 2x + 3 = 0$$

Can be simply manipulated to yield:

$$x = \frac{x^2 + 3}{2}$$

- Whereas $sinx = 0$ could be put into the form of $x = g(x)$ by adding x to both sides to yield:

$$x = sinx + x$$

Cont …

- The utility of the equation of $x = g(x)$ is that it provides a formula to predict a new value of x as a function of an old value of x.
- Thus, given an initial guess at the root x_i, the equation of $x = g(x)$ can be used to compute a new estimate x_{i+1} as expressed by the iterative formula:

$$x_{i+1} = g(x_i)$$

- As with other iterative formulas in this book, the approximate error for this equation can be determined using the error estimator:

$$\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| * 100\%$$

Example: Simple Fixed – Point Iteration

- **Problem Statement**: Use simple fixed – point iteration to locate the root of:

$$f(x) = e^{-x} - x$$

- **Solution:** The function can be separated directly and expressed in the form of $x_{i+1} = g(x_i)$ as:

$$x_{i+1} = e^{-x_i}$$

- Starting with an initial guess of $x_0 = 0$, this iterative equation can be applied to compute:

i	x_i	ε_a (%)	ε_t (%)
0	0		100.0
1	1.000000	100.0	76.3
2	0.367879	171.8	35.1
3	0.692201	46.9	22.1
4	0.500473	38.3	11.8
5	0.606244	17.4	6.89
6	0.545396	11.2	3.83
7	0.579612	5.90	2.20
8	0.560115	3.48	1.24
9	0.571143	1.93	0.705
10	0.564879	1.11	0.399

Cont …

- Thus, each iteration brings the estimate closer to the true value of the root: 0.56714329.

Convergence

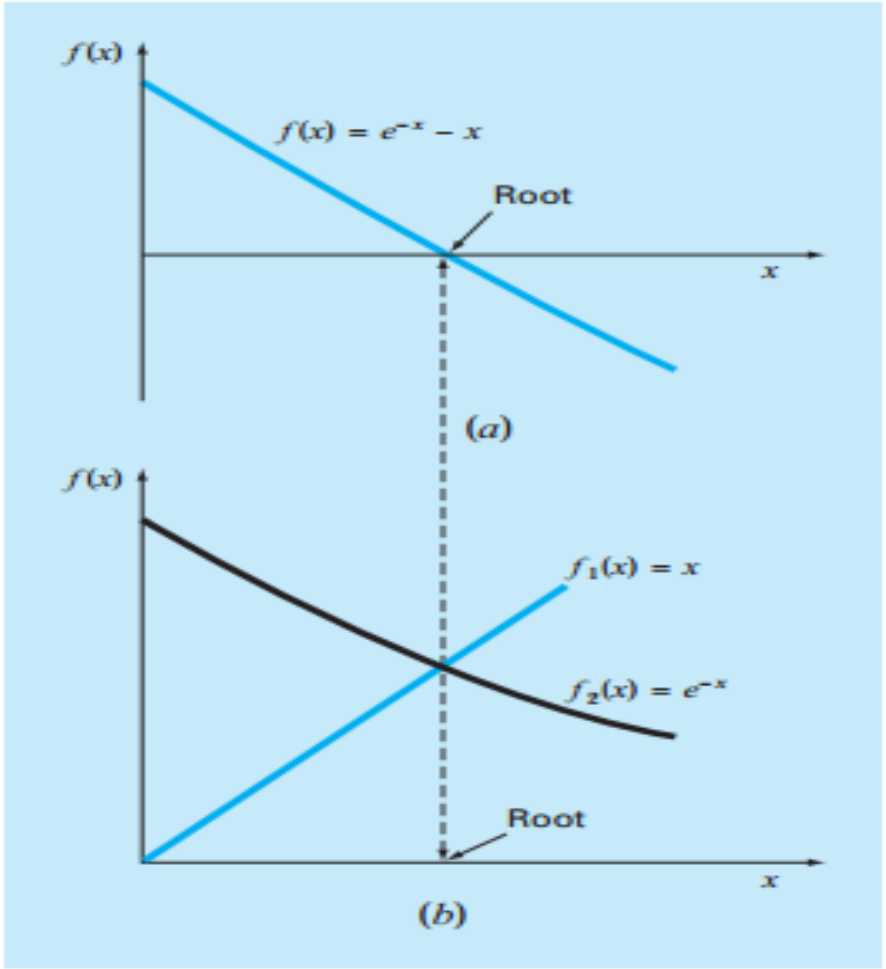
- Notice that the true percent relative error for each iteration of Example above is roughly proportional (by a factor of about 0.5 to 0.6) to the error from the previous iteration.
- This property, called **linear convergence**, is characteristic of fixed – point iteration.

Cont …

- Aside from the "rate" of convergence, we must comment at this point about the "possibility" of convergence.
- The concepts of convergence and divergence can be depicted graphically.
- Recall that, we graphed a function to visualize its structure and behavior.
- Such an approach is employed in Figure "a" below for the function:

$$f(x) = e^{-x} - x$$

FIGURE
Two alternative graphical methods for determining the root of $f(x) = e^{-x} - x$. (*a*) Root at the point where it crosses the x axis; (*b*) root at the intersection of the component functions.



Cont …

- An alternative graphical approach is to separate the equation into two component parts, as in:

$$f_1(x) = f_2(x)$$

- Then the two equations:

$$y_1 = f_1(x) \qquad and \qquad y_2 = f_2(x)$$

- Can be plotted separately (Figure "b" above).
- The x values corresponding to the intersections of these functions represent the roots of $f(x) = 0$.

EXAMPLE: The Two – Curve Graphical Method

- **Problem Statement:** Separate the equation $e^{-x} - x = 0$ into two parts and determine its root graphically.
- **Solution:** Reformulate the equation as:

$$y_1 = x \; and \; y_2 = e^{-x}$$

- The following values can be computed:

x	y_1	y_2
0.0	0.0	1.000
0.2	0.2	0.819
0.4	0.4	0.670
0.6	0.6	0.549
0.8	0.8	0.449
1.0	1.0	0.368

Cont …

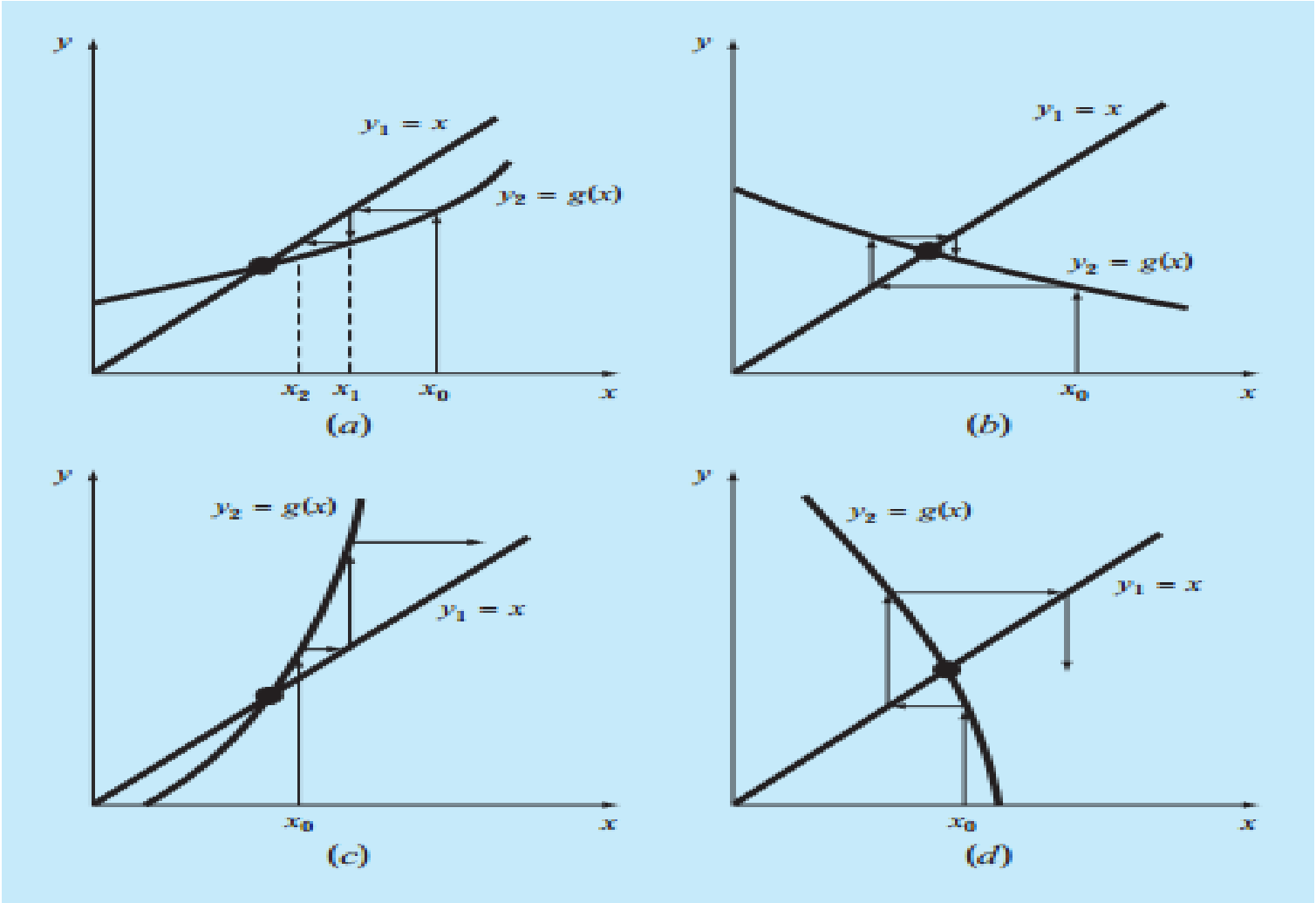
- These points are plotted in Figure "b" above.
- The intersection of the two curves indicates a root estimate of approximately $x = 0.57$, which corresponds to the point where the single curve in Figure "a" above crosses the x − axis.
- The two – curve method can now be used to illustrate the convergence and divergence of fixed – point iteration.
- First, the equation $x = g(x)$ can be re – expressed as a pair of equations

$$y_1 = x \; and \; y_2 = g(x)$$

Cont …

- These two equations can then be plotted separately.
- As was the case with $y_1 = f_1(x)\ and\ y_2 = f_2(x)$, the roots of $f(x) = 0$ correspond to the abscissa value at the intersection of the two curves.
- The function $y_1 = x$ and four different shapes for $y_2 = g(x)$ are plotted in Figure below.

FIGURE
Graphical depiction of (*a*) and (*b*) convergence and (*c*) and (*d*) divergence of simple fixed-point iteration. Graphs (*a*) and (*c*) are called monotone patterns, whereas (*b*) and (*d*) are called oscillating or spiral patterns. Note that convergence occurs when $|g'(x)| < 1$.



Cont …

- For the first case (Figure "a" above), the initial guess of x_o is used to determine the corresponding point on the y_2 curve $[x_o, g(x_o)]$.
- The point (x_1, x_1) is located by moving left horizontally to the y_1 curve.
- These movements are equivalent to the first iteration in the fixed-point method: $x_1 = g(x_o)$.
- Thus, in both the equation and in the plot, a starting value of x_o is used to obtain an estimate of x_1.
- The next iteration consists of moving to $[x_1, g(x_1)]$ and then to (x_2, x_2).

Cont …

- This iteration is equivalent to the equation:

$$x_2 = g(x_1)$$

- The solution in Figure "a" is convergent because the estimates of x move closer to the root with each iteration.
- The same is true for Figure "b".
- However, this is not the case for Figure "c" and "d", where the iterations diverge from the root.

Cont …

- Notice that convergence seems to occur only when the absolute value of the slope of $y_2 = g(x)$ is less than the slope of $y_1 = x$, that is, when:

$$|g'(x)| < 1$$

(On your text book Box 6.1 provides a theoretical derivation of this result.)

The NEWTON – RAPHSON Method

- Perhaps the most widely used of all root-locating formulas is the Newton-Raphson equation (Figure below).
- If the initial guess at the root is x_i, a tangent can be extended from the point $[x_i, f(x_i)]$.
- The point where this tangent crosses the x axis usually represents an improved estimate of the root.
- The Newton-Raphson method can be derived on the basis of this geometrical interpretation (an alternative method based on the Taylor series is described in Box 6.2 in your text book).

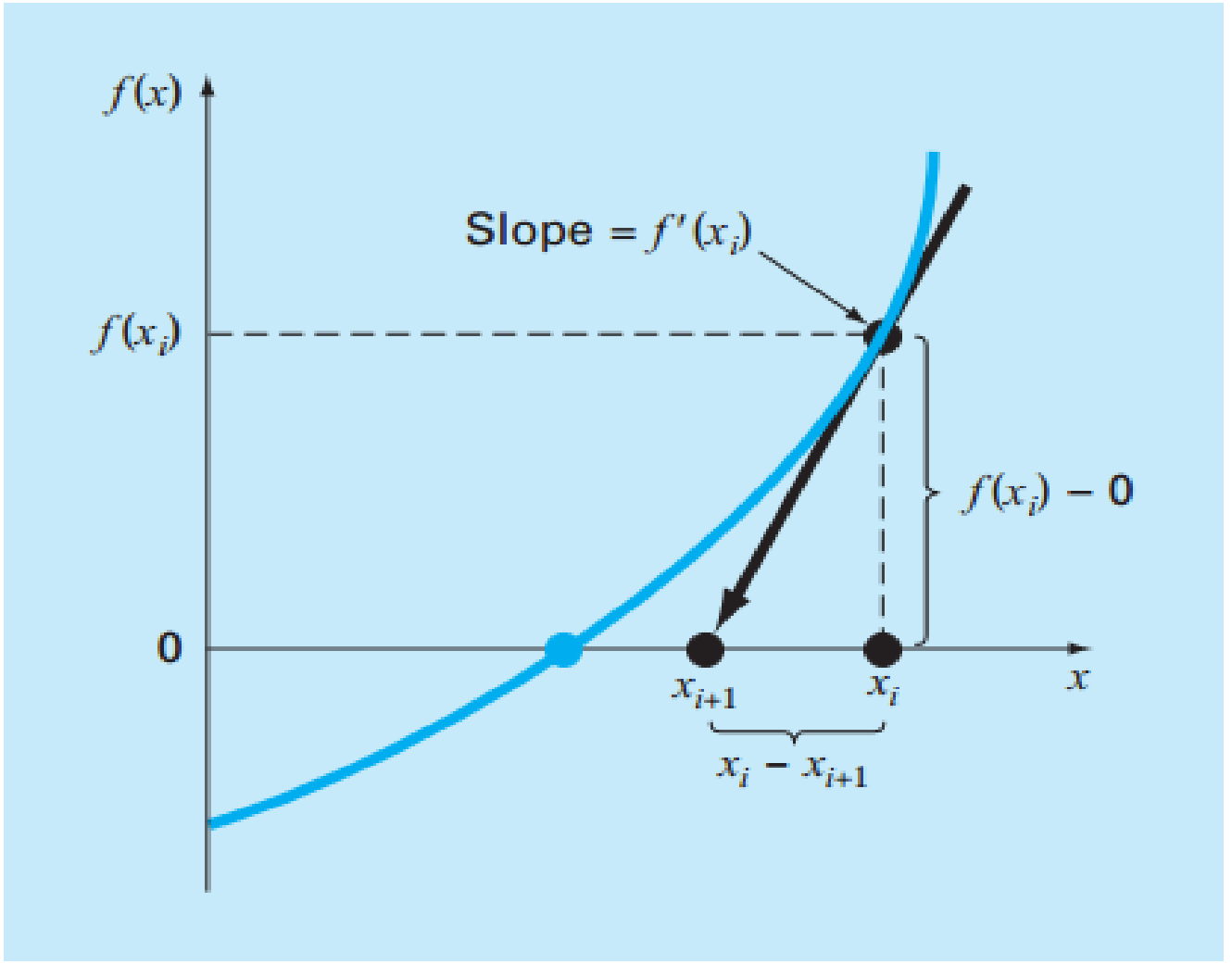


FIGURE
Graphical depiction of the Newton-Raphson method. A tangent to the function of x_i [that is, $f'(x_i)$] is extrapolated down to the x axis to provide an estimate of the root at x_{i+1}.

Cont …

- As in Figure above, the first derivative at x is equivalent to the slope:

$$f'(x_i) = \frac{f(x_i) - 0}{x_i - x_{i+1}}$$

- Which can be rearranged to yield:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

- Which is called the **Newton-Raphson** formula.

EXAMPLE: Newton - Raphson Method

Problem Statement: Use the Newton-Raphson method to estimate the root of:

$$f(x) = e^{-x} - x$$

- Employing an initial guess of $x_o = 0$.

Solution: The first derivative of the function can be evaluated as:

$$f'(x) = -e^{-x} - 1$$

- Which can be substituted along with the original function into Newton Raphson

Cont …

- Equation to give:

$$x_{i+1} = x_i - \frac{e^{-x} - x}{-e^{-x} - 1}$$

- Starting with an initial guess of $x_o = 0$, this iterative equation can be applied to compute:
- Thus, the approach rapidly converges on the true root.
- Notice that the true percent relative error at each iteration decreases much faster than it does in simple fixed-point iteration.

i	x_i	ε_t (%)
0	0	100
1	0.500000000	11.8
2	0.566311003	0.147
3	0.567143165	0.0000220
4	0.567143290	$< 10^{-8}$

Newton – Raphson formula using a Taylor series

- Aside from the geometric derivation of Newton – Raphson Equations above, it may also be developed from the Taylor series expansion.
- This alternative derivation is useful in that it also provides insight into the rate of convergence of the method.
- Taylor series expansion can be represented as:

$$f(x_{i+1}) = f(x_i) + f'(x_i)(x_{i+1} - x_i) + \frac{f''(\xi)}{2!}(x_{i+1} - x_i)^2$$

- Where ξ lies somewhere in the interval from $x_i \; to \; x_{i+1}$.

Cont …

- An approximate version is obtainable by truncating the series after the first derivative term:

$$f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1} - x_i)$$

- At the intersection with the $x - axis$, $f(x_{i+1})$ would be equal to zero, or

$$0 = f(x_i) + f'(x_i)(x_{i+1} - x_i)$$

- Which can be solved for:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Cont …

- Which is identical to Newton Raphson Formula derived earlier.
- Thus, we have derived the Newton Raphson formula using a Taylor series.
- Aside from the derivation, the Taylor series can also be used to estimate the error of the formula.
- This can be done by realizing that if the complete Taylor series were employed, an exact result would be obtained.
- For this situation $x_{i+1} = x_r$, where x is the true value of the root.

Cont …

- Substituting this value along with $f(x_r) = 0$ into Taylor series expansion given above yields:

$$0 = f(x_i) + f'(x_i)(x_r - x_i) + \frac{f''(\xi)}{2!}(x_r - x_i)^2$$

- An approximation equation:

$$0 = f(x_i) + f'(x_i)(x_{i+1} - x_i)$$

- Can be subtracted from the above equation to give:

$$0 = f'(x_i)(x_r - x_{i+1}) + \frac{f''(\xi)}{2!}(x_r - x_i)^2$$

Cont …

- Now, realize that the error is equal to the discrepancy between x_{i+1} and the true value x_r, as in:

$$E_{t,i+1} = x_r - x_{i+1}$$

- Hence, the equation can be expressed as:

$$0 = f'(x_i)E_{t,i+1} + \frac{f''(\xi)}{2!}{E_{t,i}}^2$$

- If we assume convergence, both x_i and ξ should eventually be approximated by the root x_r, and the equation can be rearranged to yield:

$$E_{t,i+1} = \frac{-f''(x_r)}{2f'(x_r)}{E_{t,i}}^2$$

Cont …

- According to this Equation, the error is roughly proportional to the square of the previous error.
- This means that the number of correct decimal places approximately doubles with each iteration.
- Such behavior is referred to as **quadratic** convergence.
- The following Example manifests this property.

EXAMPLE: Error Analysis of Newton - Raphson Method

- **Problem Statement:** As derived earlier, the Newton – Raphson method is quadratically convergent. That is, the error is roughly proportional to the square of the previous error, as in:

$$E_{t,i+1} = \frac{-f''(x_r)}{2f'(x_r)} {E_{t,i}}^2$$

- Examine this formula and see if it applies to the results of the previous Example.

Cont …

- **Solution:** The first derivative of $f(x) = e^{-x} - x$ is:

$$f'(x) = -e^{-x} - 1$$

- Which can be evaluated at $x_r = 0.56714329$ as:

$$f'(x_r) = f'(0.56714329) = -1.56714329$$

- The second derivative is:

$$f''(x) = -e^{-x}$$

- Which can be evaluated as:

$$f''(x_r) = f''(0.56714329) = 0.56714329$$

Cont …

- These results can be substituted into the Equation to yield:

$$E_{t,i+1} \cong \frac{-f''(x_r)}{2f'(x_r)} {E_{t,i}}^2 = \frac{-0.56714329}{2(-1.56714329)} {E_{t,i}}^2 = 0.18095 {E_{t,i}}^2$$

- From previous Example, the initial error was $E_{t,0} = 0.56714329$, which can be substituted into the error equation to predict:

$$E_{t,1} \cong 0.18095 {E_{t,0}}^2 = 0.18095(0.56714329)^2 = 0.0582$$

- Which is close to the true error of 0.06714329.

Cont …

- **For the next iteration,**

$$E_{t,2} \cong 0.18095 {E_{t,1}}^2 = 0.18095(0.06714329)^2 = 0.0008158$$

Which also compares favorably with the true error of 0.0008323.

- **For the third iteration**

$$E_{t,3} \cong 0.18095 {E_{t,2}}^2 = 0.18095(0.0008323)^2 = 0.000000125$$

- Which is the error obtained in previous Example.
- The error estimate improves in this manner because, as we come closer to the root, x and ξ are better approximated by x_r.

Cont …

- Finally,

$$E_{t,4} \cong 0.18095 {E_{t,3}}^2 = 0.18095(0.000000125)^2 = 2.83 * 10^{-15}$$

- Thus, this example illustrates that the error of the Newton – Raphson method for this case is, in fact, roughly proportional (by a factor of 0.18095) to the square of the error of the previous iteration.

Pitfalls of the Newton – Raphson Method

- Although the Newton – Raphson method is often very efficient, there are situations where it performs poorly.
- A special case—multiple roots—will be addressed later in this chapter.
- However, even when dealing with simple roots, difficulties can also arise, as in the following example.
- **EXAMPLE: Example of a Slowly Converging Function with Newton-Raphson**
- **Problem Statement:** Determine the positive root of $f(x) = x^{10} - 1$ using the Newton – Raphson method and an initial guess of $x = 0.5$.

Cont …

- **Solution:** The Newton-Raphson formula for this case is:

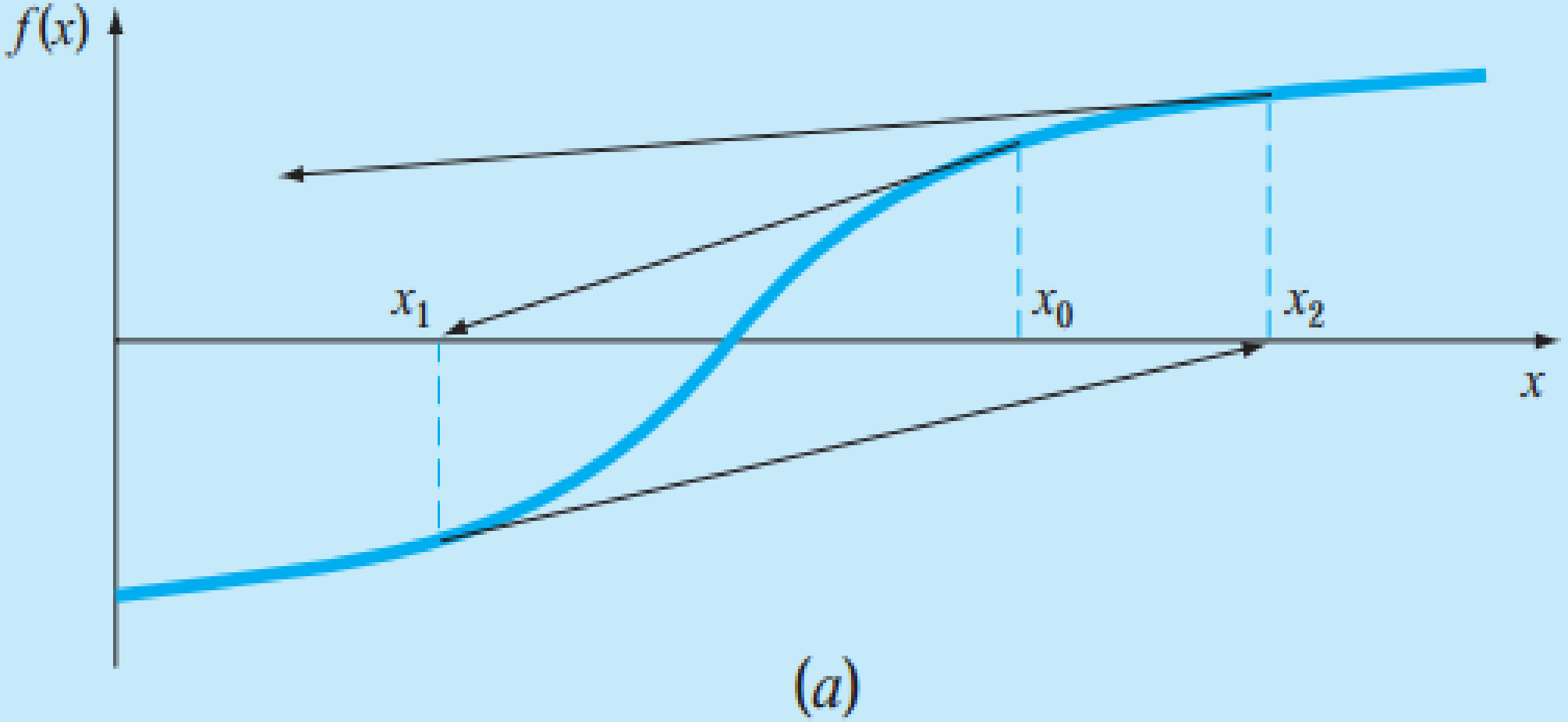
$$x_{i+1} = x_i - \frac{x_i^{10} - 1}{10x_i^9}$$

- Which can be used to compute:
- Thus, after the first poor prediction, the technique is converging on the true root of 1, but at a very slow rate.

Iteration	x
0	0.5
1	51.65
2	46.485
3	41.8365
4	37.65285
5	33.887565
.	
.	
.	
∞	1.0000000

Cont …

- Aside from slow convergence due to the nature of the function, other difficulties can arise, as illustrated in Figure below (Four cases where the Newton-Raphson method exhibits poor convergence).
- For example;
  - Figure "a" depicts the case where an inflection point [that is, $f''(x) = 0$] occurs in the vicinity of a root. Notice that iterations beginning at x_0 progressively diverge from the root.

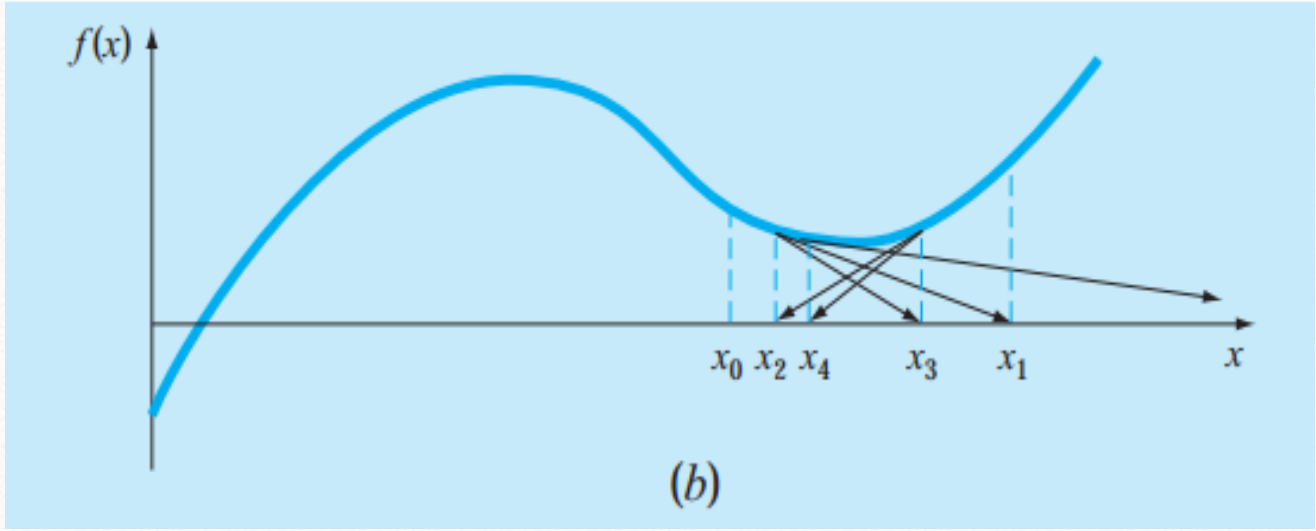


(a)

Cont …

- Figure “b” illustrates the tendency of the Newton – Raphson technique to oscillate around a local maximum or minimum.

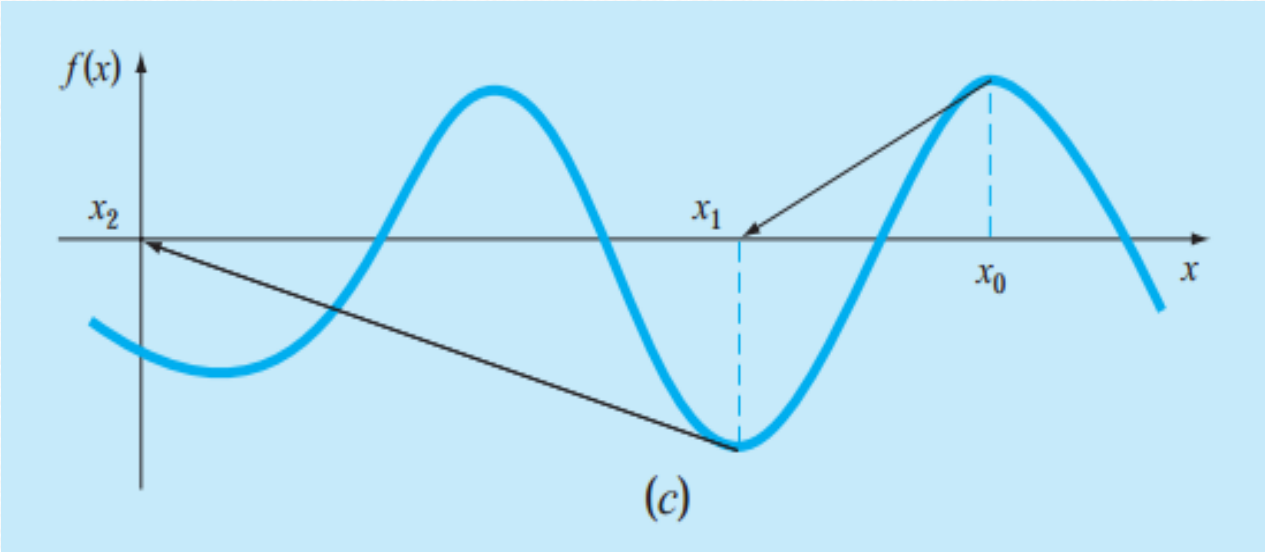
- Such oscillations may persist, or as in Figure “b”, a near-zero slope is reached, where up on the solution is sent far from the area of interest.



Cont …

- Figure “c” shows how an initial guess that is close to one root can jump to a location several roots away.

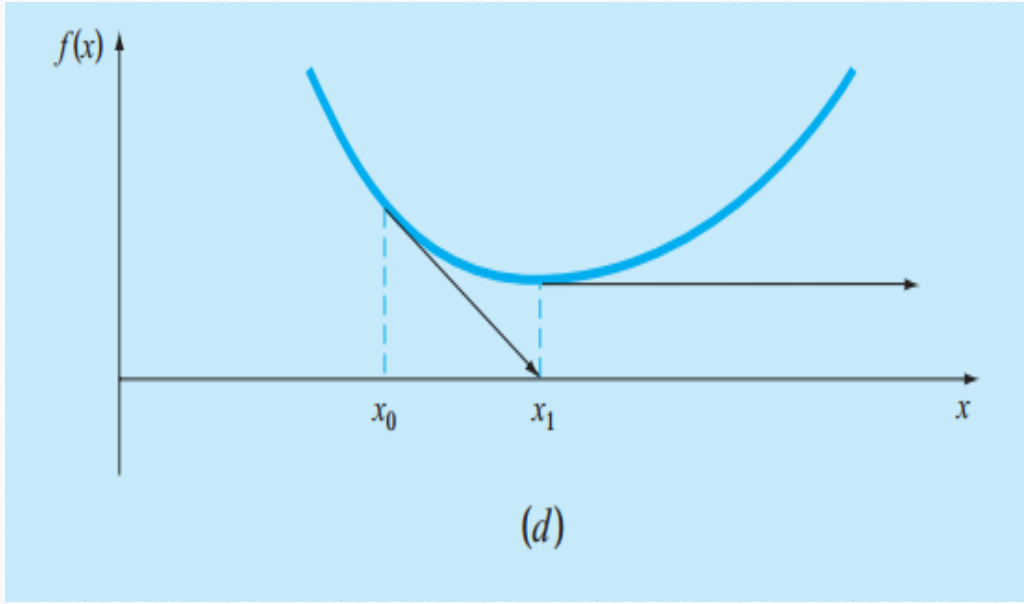
- This tendency to move away from the area of interest is because near zero slopes are encountered.



Cont …

- Obviously, a zero slope $[f'(x) = 0]$ is truly a disaster because it causes division by zero in the Newton-Raphson formula.

- Graphically (see Figure "d"), it means that the solution shoots off horizontally and never hits the x axis.
- Thus, there is no general convergence criterion for Newton-Raphson.
- Its convergence depends on the nature of the function and on the accuracy of the initial guess.



(d)

Cont …

- The only remedy is to have an initial guess that is "sufficiently" close to the root.
- And for some functions, no guess will work! Good guesses are usually predicated on knowledge of the physical problem setting or on devices such as graphs that provide insight into the behavior of the solution.
- The lack of a general convergence criterion also suggests that good computer software should be designed to recognize slow convergence or divergence.

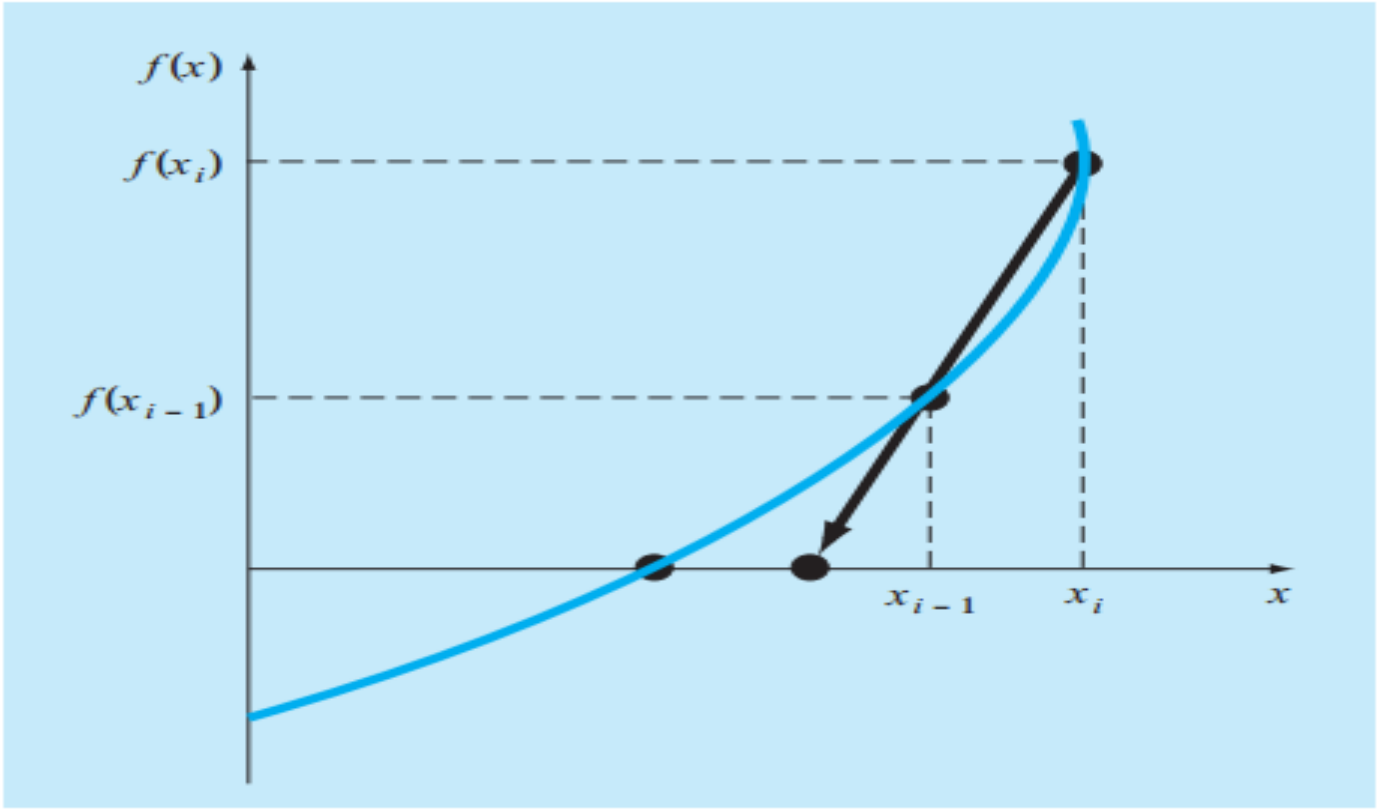
The SECANT Method

- A potential problem in implementing the Newton – Raphson method is the evaluation of the derivative.
- Although this is not inconvenient for polynomials and many other functions, there are certain functions whose derivatives may be extremely difficult or inconvenient to evaluate.
- For these cases, the derivative can be approximated by a backward finite divided difference, as in (Figure below)

$$f'(x_i) \cong \frac{f(x_{i-1}) - f(x_i)}{x_{i-1} - x_i}$$

FIGURE

Graphical depiction of the secant method. This technique is similar to the Newton-Raphson technique in the sense that an estimate of the root is predicted by extrapolating a tangent of the function to the x axis. However, the secant method uses a difference rather than a derivative to estimate the slope.



Cont …

- This approximation can be substituted into Newton – Raphson Equation:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

- To yield the following iterative equation:

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

- This Equation is the formula for the **secant method**.

Cont …

- Notice that the approach requires two initial estimates of x.
- However, because f(x) is not required to change signs between the estimates, it is not classified as a bracketing method.

EXAMPLE: The Secant Method

- **Problem Statement:** Use the secant method to estimate the root of:

$$f(x) = e^{-x} - x$$

Start with initial estimates of $x_{-1} = 0 \; and \; x_0 = 1.0$

Cont …

Solution: Recall that the true root is 0.56714329. . . .

- First iteration:

$$x_{-1} = 0 \qquad f(x_{-1}) = 1.00000$$

$$x_0 = 1 \qquad f(x_0) = -0.63212$$

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

$$x_1 = x_0 - \frac{f(x_0)(x_{-1} - x_0)}{f(x_{-1}) - f(x_0)} = 1 - \frac{-0.63212(0 - 1)}{1 - (-0.63212)} = 0.61270$$

$$\varepsilon_t = 8.0\%$$

Cont …

- Second iteration:

$x_0 = 1$	$f(x_0) = -0.63212$
$x_1 = 0.61270$	$f(x_1) = -0.07081$

(Note that both estimates are now on the same side of the root.)

$$x_2 = x_1 - \frac{f(x_1)(x_0 - x_1)}{f(x_0) - f(x_1)} = 0.61270 - \frac{-0.07081(1 - 0.61270)}{-0.63212 - (-0.07081)}$$

$$= 0.56384$$

$$\varepsilon_t = 0.58\%$$

Cont …

- Third iteration:

$$x_1 = 0.61270 \qquad f(x_1) = -0.07081$$

$$x_2 = 0.56384 \qquad f(x_2) = 0.00518$$

$$x_3 = x_2 - \frac{f(x_2)(x_1 - x_2)}{f(x_1) - f(x_2)}$$

$$= 0.56384 - \frac{0.00518(0.61270 - 0.56384)}{-0.07081 - 0.00518}$$

$$= 0.56717$$

$$\varepsilon_t = 0.0048\%$$

The Difference between the Secant and False-Position Methods

- Note the similarity between the secant method and the false-position method. For example, Equation for the secant method:

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

- And Equation for False Position method:

$$x_r = x_u - \frac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$$

are identical on a term-by-term basis.

Cont …

- Both use two initial estimates to compute an approximation of the slope of the function that is used to project to the x axis for a new estimate of the root.
- However, a critical difference between the methods is how one of the initial values is replaced by the new estimate.
- Recall that in the **false-position method** the latest estimate of the root replaces whichever of the original values yielded a function value with the same sign as $f(x_r)$.
- Consequently, the two estimates always bracket the root.

Cont …

- Therefore, for all practical purposes, the method always converges because the root is kept within the bracket.

- In contrast, the secant method replaces the values in strict sequence, with the new value x_{i+1} replacing x_i and x_i replacing x_{i-1}.
- As a result, the two values can sometimes lie on the same side of the root.
- For certain cases, this can lead to divergence.

EXAMPLE: Comparison of Convergence of the Secant and False-Position Techniques

- **Problem Statement:** Use the false-position and secant methods to estimate the root of $f(x) = \ln x$.
- Start the computation with values of $x_l = x_{i-1} = 0.5 \; and \; x_u = x_i = 5.0$.
- **Solution:** For the false-position method, the use of False – position equation and the bracketing criterion for replacing estimates results in the following iterations:

Cont …

Iteration	x_l	x_u	x_r
1	0.5	5.0	1.8546
2	0.5	1.8546	1.2163
3	0.5	1.2163	1.0585

- As can be seen in figure "a" and "c" below, the estimates are converging on the true root which is equal to 1.
- For the secant method, using

  Equation for secant method and

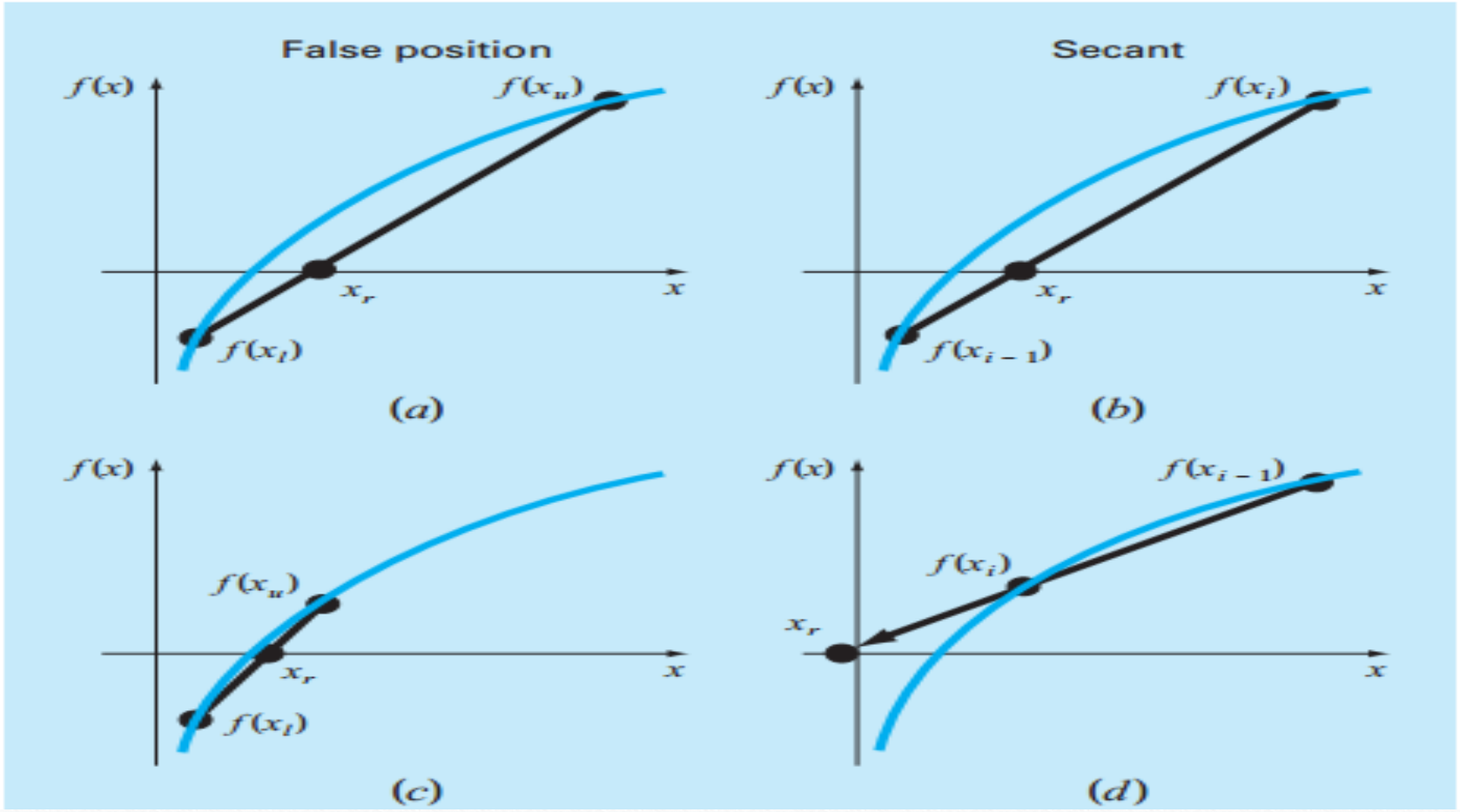
  the sequential criterion for replacing

  estimates results in:

Iteration	x_{i-1}	x_i	x_{i+1}
1	0.5	5.0	1.8546
2	5.0	1.8546	−0.10438

- As in Figure "d", the approach is divergent.

Comparison of the false-position and the secant methods. The first iterations (*a*) and (*b*) for both techniques are identical. However, for the second iterations (*c*) and (*d*), the points used differ. As a consequence, the secant method can diverge, as indicated in (*d*).



Cont …

- Although the secant method may be divergent, when it converges it usually does so at a quicker rate than the false-position method.
- For instance, the following Figure demonstrates the superiority of the secant method in this regard.
- The inferiority of the false position method is due to one end staying fixed to maintain the bracketing of the root.
- This property, which is an advantage in that it prevents divergence, is a shortcoming with regard to the rate of convergence; it makes the finite-difference estimate a less-accurate approximation of the derivative.

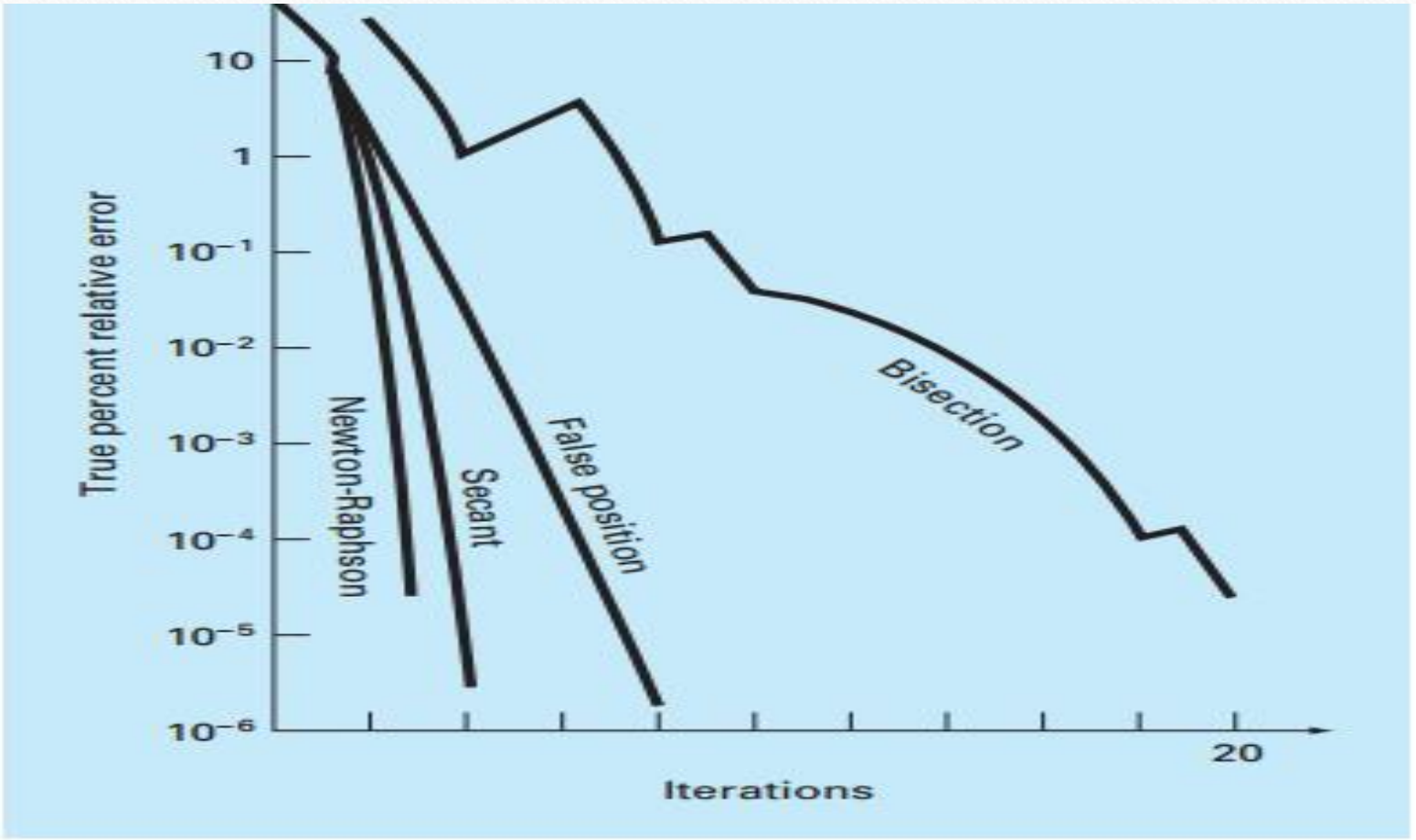


FIGURE
Comparison of the true percent relative errors ε_t for the methods to determine the roots of $f(x) = e^{-x} - x$.

Modified Secant Method

- Rather than using two arbitrary values to estimate the derivative, an alternative approach involves a fractional perturbation/alarm of the independent variable to estimate $f(x)$,

$$f'(x_i) \cong \frac{f(x_i + \delta x_i) - f(x_i)}{\delta x_i}$$

- Where δ = a small perturbation fraction.
- This approximation can be substituted into Equation:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Cont …

- To yield the following iterative equation:

$$x_{i+1} = x_i - \frac{\delta x_i f(x_i)}{f(x_i + \delta x_i) - f(x_i)}$$

EXAMPLE: Modified Secant Method

- **Problem Statement:** Use the modified secant method to estimate the root of

$$f(x) = e^{-x} - x$$

- Use a value of 0.01 for δ and start with $x_0 = 1.0$.
- Recall that the true root is 0.56714329. . . .

Cont …

- First iteration:

$$x_0 = 1.0 \qquad f(x_0) = -0.63212$$

$$x_0 + \delta x_0 = 1.01 \qquad f(x_0 + \delta x_0) = -0.64578$$

$$x_{i+1} = x_i - \frac{\delta x_i f(x_i)}{f(x_i + \delta x_i) - f(x_i)}$$

$$x_1 = x_0 - \frac{\delta x_0 f(x_0)}{f(x_0 + \delta x_0) - f(x_0)} = 1 - \frac{0.01 * 1.0 * (-0.63212)}{-0.64578 - (-0.63212)}$$

$$= 0.537263$$

$$\varepsilon_t = 5.3\%$$

Cont …

- Second iteration:

$$x_1 = 0.537263 \qquad f(x_1) = 0.047083$$

$$x_1 + \delta x_1 = 0.542635 \qquad f(x_1 + \delta x_1) = 0.038579$$

$$x_2 = 0.537263 - \frac{0.005373(0.047083)}{0.038579 - 0.047083} = 0.56701$$

$$\varepsilon_t = 0.0236\%$$

- Third iteration:

$$x_2 = 0.56701 \qquad f(x_2) = 0.000209$$

$$x_2 + \delta x_2 = 0.572680 \qquad f(x_2 + \delta x_2) = -0.00867$$

Cont …

$$x_3 = 0.56701 - \frac{0.00567(0.000209)}{-0.00867 - 0.000209} = 0.567143$$

$$\varepsilon_t = 2.365 * 10^{-5}\%$$

- The choice of a proper value for δ is not automatic:
  - If δ is too small, the method can be swamped/busy by round-off error caused by subtractive cancellation in the denominator of the Equation.
  - If it is too big, the technique can become inefficient and even divergent.
  - However, if chosen correctly, it provides a nice alternative for cases where evaluating the derivative is difficult and developing two initial guesses is inconvenient

MULTIPLE ROOTS

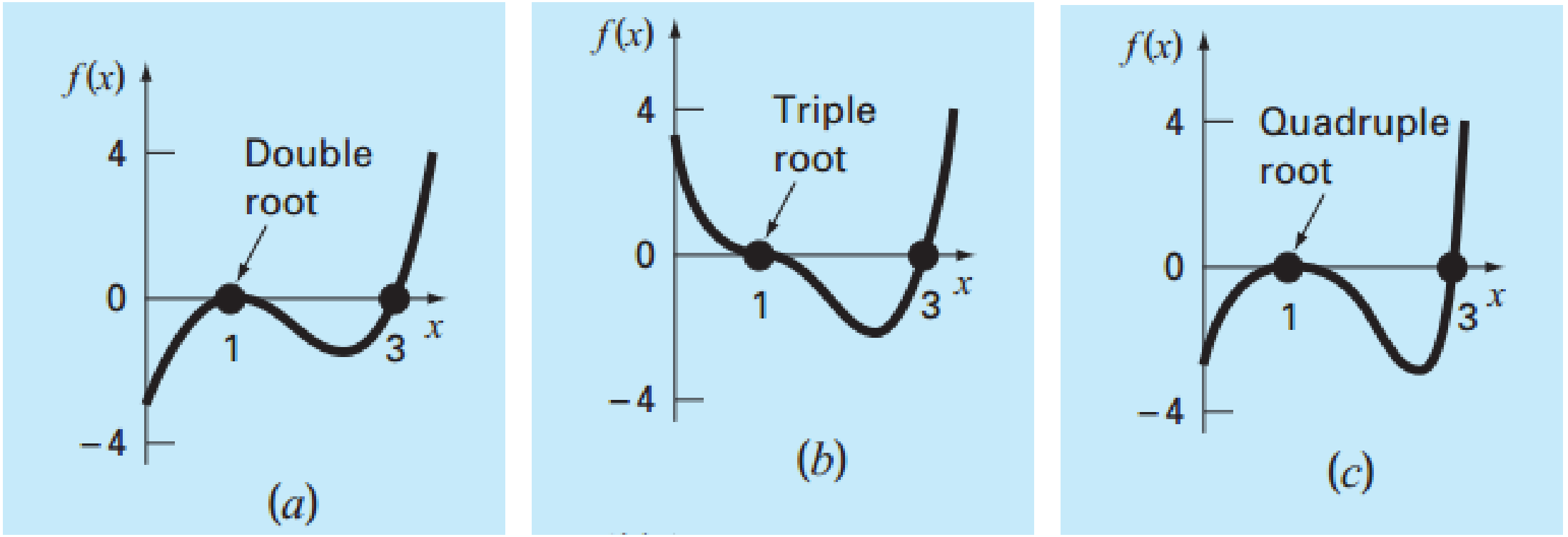
- A multiple root corresponds to a point where a function is tangent to the x axis. For example, a double root results from:

$$f(x) = (x-3)(x-1)(x-1)$$

- or, multiplying terms:

$$f(x) = x^3 - 5x^2 + 7x - 3$$

- The equation has a double root because one value of x makes two terms in Equation above equal to zero.
- Graphically, this corresponds to the curve touching the x – axis tangentially at the double root.
- Examine Figure "a" below at x = 1.



FIGURE

Examples of multiple roots that are tangential to the x axis. Notice that the function does not cross the axis on either side of even multiple roots (*a*) and (*c*), whereas it crosses the axis for odd cases (*b*).

Cont …

- Notice that the function touches the axis but does not cross it at the root.
- A triple root corresponds to the case where one x value makes three terms in an equation equal to zero, as in:

$$f(x) = (x-3)(x-1)(x-1)(x-1)$$

- Or, multiplying terms,

$$f(x) = x^4 - 6x^3 + 12x^2 - 10x + 3$$

- Notice that the graphical depiction (Figure "b") again indicates that the function is tangent to the axis at the root, but that for this case the axis is crossed.

Cont …

- In general, odd multiple roots cross the axis, whereas even ones do not. For example, the quadruple root in Figure "c" does not cross the axis.
- Multiple roots pose some difficulties for many of the numerical methods described in Part Two:
  1) The fact that the function does not change sign at even multiple roots precludes the use of the reliable bracketing methods that were discussed earlier.
  - Thus, of the methods covered in this book, you are limited to the open methods that may diverge.

Cont …

2) Another possible problem is related to the fact that not only $f(x)$ but also $f'(x)$ goes to zero at the root.

- This poses problems for both the Newton – Raphson and secant methods, which both contain the derivative (or its estimate) in the denominator of their respective formulas.
- This could result in division by zero when the solution converges very close to the root.

- A simple way to avoid these problems is based on the fact that it can be demonstrated theoretically (Ralston and Rabinowitz, 1978) that $f(x)$ will always reach zero before $f'(x)$.

Cont …

- Therefore, if a zero check for f (x) is incorporated into the computer program, the computation can be terminated before $f'(x)$ reaches zero.

  3) It can be demonstrated that the Newton – Raphson and secant methods are linearly, rather than quadratically, convergent for multiple roots (Ralston and Rabinowitz, 1978).

  - Modifications have been proposed to alleviate this problem.

- Ralston and Rabinowitz (1978) have indicated that a slight change in the formulation returns it to quadratic convergence, as in:

$$x_{i+1} = x_i - m\frac{f(x_i)}{f'(x_i)}$$

Cont …

- Where "m" is the multiplicity of the root (that is, m = 2 for a double root, m = 3 for a triple root, etc.).
- Of course, this may be an unsatisfactory alternative because it hinges on foreknowledge of the multiplicity of the root.
- Another alternative, also suggested by Ralston and Rabinowitz (1978), is to define a new function u(x), that is, the ratio of the function to its derivative, as in:

$$u(x) = \frac{f(x)}{f'(x)}$$

Cont …

- It can be shown that this function has roots at all the same locations as the original function.
- Therefore, this equation can be substituted into the equation:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

- To develop an alternative form of the Newton – Raphson method:

$$x_{i+1} = x_i - \frac{u(x_i)}{u'(x_i)}$$

Cont …

- $u(x)$ Can be differentiated to give:

$$u'(x) = \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2}$$

- Substituting Equations for $u(x_i)\ and\ u'(x_i)$ can be substituted into Equation x_{i+1} the result simplified to yield:

$$x_{i+1} = x_i - \frac{f(x_i)f'(x_i)}{[f'(x_i)]^2 - f(x_i)f''(x_i)}$$

EXAMPLE: Modified Newton – Raphson Method for Multiple Roots

- **Problem Statement:** Use both the standard and modified Newton-Raphson methods to evaluate the multiple root of:

$$f(x) = (x-3)(x-1)(x-1) = x^3 - 5x^2 + 7x - 3$$

- With an initial guess of $x_0 = 0$.

Solution: The first derivative of $f(x)$ is:

$$f'(x) = 3x^2 - 10x + 7$$

- And therefore, the standard Newton-Raphson method for this problem is:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} = x_i - \frac{{x_i}^3 - 5{x_i}^2 + 7x_i - 3}{3{x_i}^2 - 10x_i + 7}$$

Cont …

- Which can be solved iteratively for:
- As anticipated, the method is linearly convergent toward the true value of 1.0.
- For the modified method, the second derivative is:

$$f''(x) = 6x - 10$$

i	x_i	ε_t (%)
0	0	100
1	0.4285714	57
2	0.6857143	31
3	0.8328654	17
4	0.9133290	8.7
5	0.9557833	4.4
6	0.9776551	2.2

Cont …

- And the iterative relationship is:

$$x_{i+1} = x_i - \frac{f(x_i)f'(x_i)}{[f'(x_i)]^2 - f(x_i)f''(x_i)}$$

$$= x_i - \frac{({x_i}^3 - 5{x_i}^2 + 7x_i - 3)(3{x_i}^2 - 10x_i + 7)}{[3{x_i}^2 - 10x_i + 7]^2 - ({x_i}^3 - 5{x_i}^2 + 7x_i - 3)(6x_i - 10)}$$

Which can be solved for:

i	x_i	ε_t (%)
0	0	100
1	1.105263	11
2	1.003082	0.31
3	1.000002	0.00024

Cont …

- Thus, the modified formula is quadratically convergent.
- We can also use both methods to search for the single root at x = 3.
- Using an initial guess of $x_0 = 4$ gives the following results:

i	Standard	ε_t (%)	Modified	ε_t (%)
0	4	33	4	33
1	3.4	13	2.636364	12
2	3.1	3.3	2.820225	6.0
3	3.008696	0.29	2.961728	1.3
4	3.000075	0.0025	2.998479	0.051
5	3.000000	2×10^{-7}	2.999998	7.7×10^{-5}

- Thus, both methods converge quickly, with the standard method being somewhat more efficient.

Cont …

- The preceding/previous example illustrates the trade-offs involved in selecting for the modified Newton – Raphson method.
- Although it is preferable for multiple roots, it is somewhat less efficient and requires more computational effort than the standard method for simple roots.
- It should be noted that a modified version of the secant method suited for multiple roots can also be developed by substituting Equation:

$$u(x) = \frac{f(x)}{f'(x)}$$

Cont …

- Into Equation:

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

- The resulting formula is (Ralston and Rabinowitz, 1978):

$$x_{i+1} = x_i - \frac{u(x_i)(x_{i-1} - x_i)}{u(x_{i-1}) - u(x_i)}$$

END OF CHAPTER TWO

THANK YOU !!!